

MODULE 6

NULL HYPOTHESIS SIGNIFICANCE TESTING

NULL HYPOTHESIS SIGNIFICANCE TESTING

- Null hypothesis significance testing (NHST) is a procedure that uses a sample of data to determine whether an effect or association might be present at the population level
- A cornerstone of psychological research for over 100 years
- Provides a formal recipe to evaluate evidence from data, and the same steps apply to virtually any research question

SIGNIFICANCE TESTING STEPS

1

Specify hypotheses

2

Define standard of evidence

3

Design study and collect data

4

Compare data to null hypothesis

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Evaluate hypotheses and draw conclusion

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R analysis

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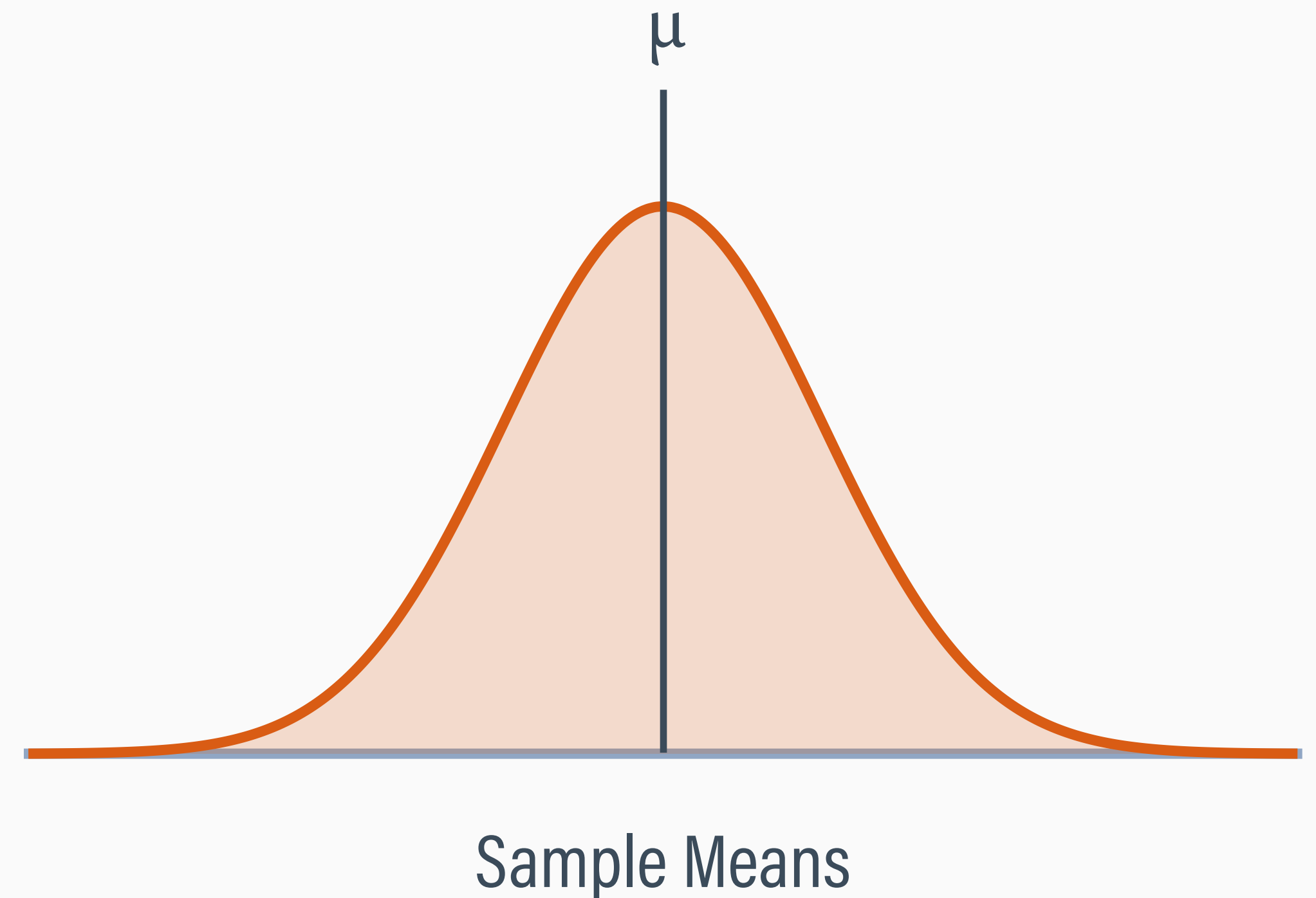
R analysis

QUICK REVIEW: SAMPLING ERROR

- The frequentist paradigm imagines a single population that spawns many hypothetical random samples of data (one parameter, many hypothetical estimates)
- The amount by which an estimate differs from the true population statistic is called sampling error
- Every hypothetical sample has a different amount of error

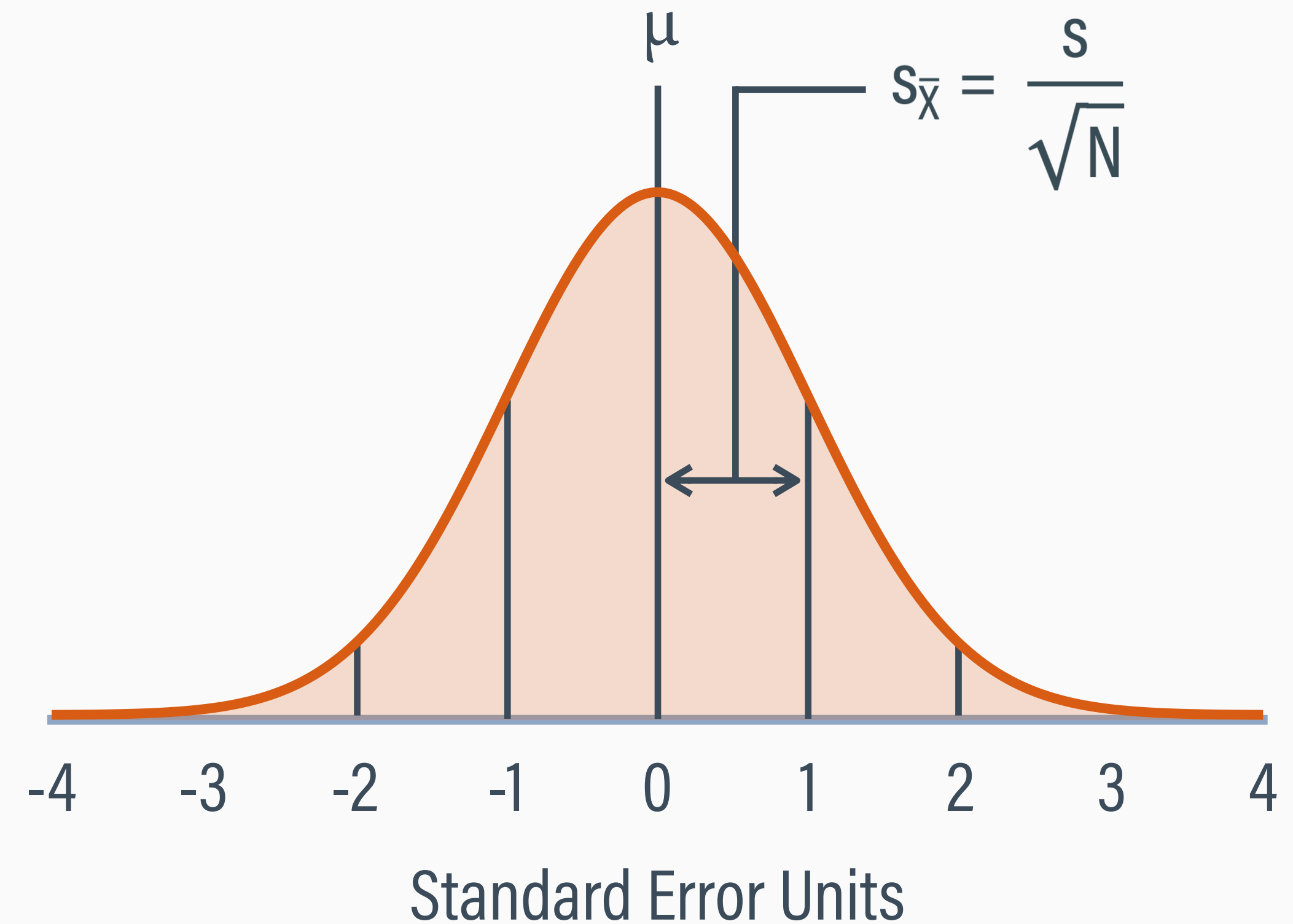
QUICK REVIEW: SAMPLING DISTRIBUTION

- The distribution of the estimates from many hypothetical samples is a sampling distribution
- With a large enough N , sample means follow a normal curve centered at the true mean
- Most estimates have small sampling errors, but a few have larger errors



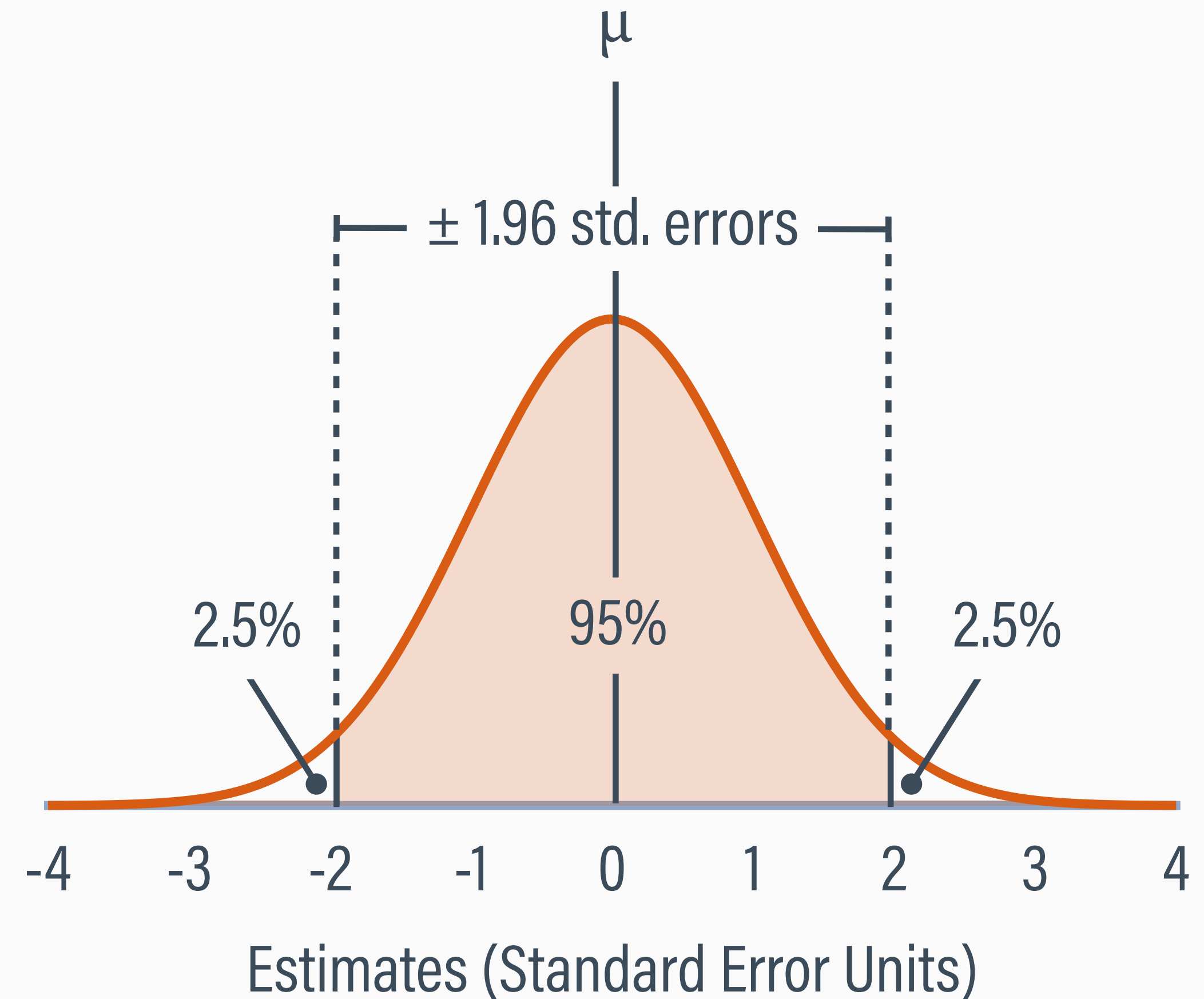
QUICK REVIEW: STANDARD ERROR

- The standard error is the average distance from a sample mean and the true mean
- $s_{\bar{x}}$ = standard deviation of the sample means
- The standard error is the average or expected amount of sampling error across many hypothetical samples



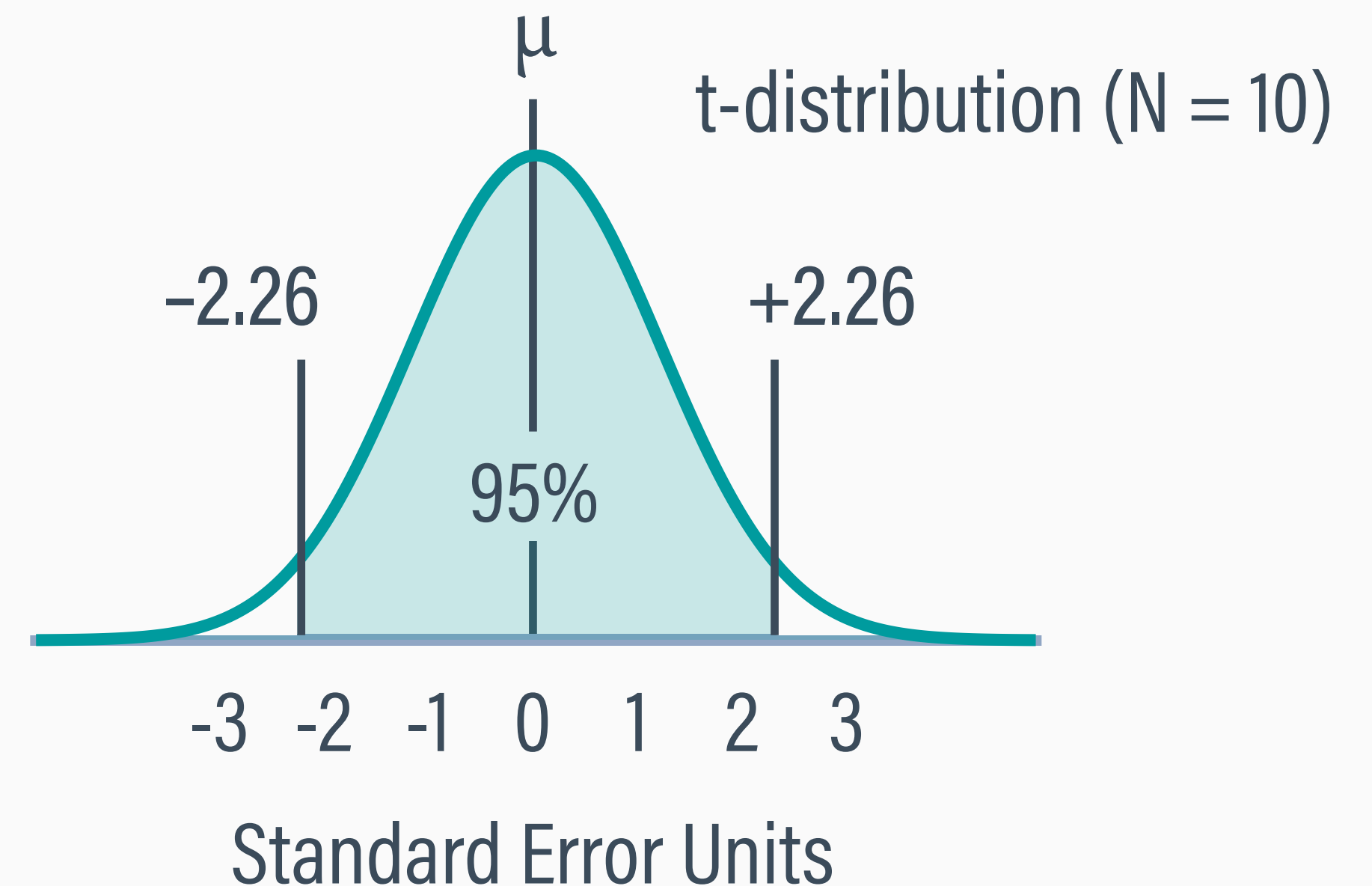
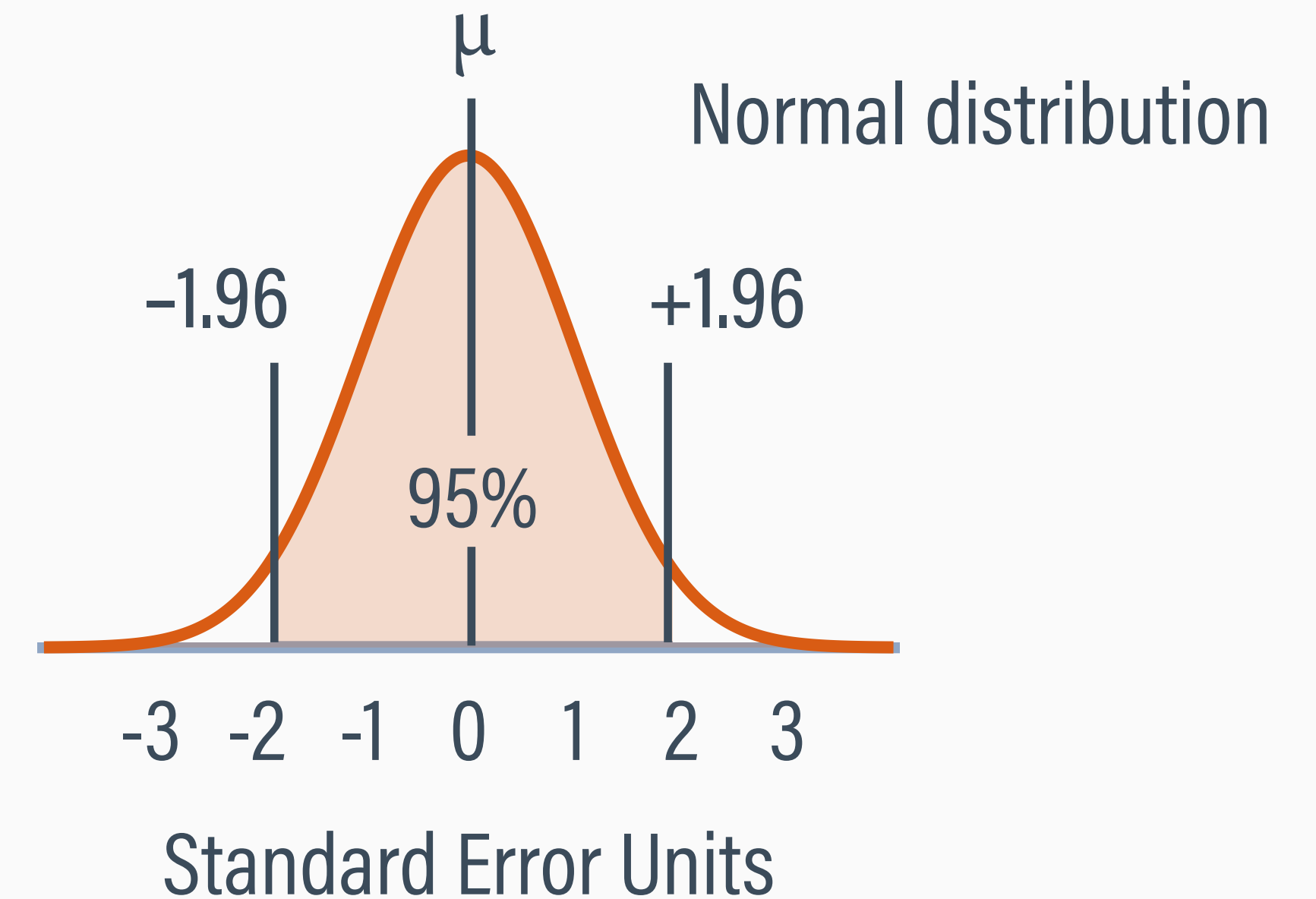
QUICK REVIEW: NORMAL CURVE RULE

- The standard error is the standard deviation of many hypothetical sample means
- We can apply normal curve rules of thumb
- 95% of the means from large samples are within ± 1.96 standard errors of the true mean



QUICK REVIEW: T-DISTRIBUTION

- When using small samples, the normal curve is an inaccurate description of sampling error
- The t-distribution is a series of bell-shaped curves that stretch out (become more variable) as the N decreases
- Small samples are more likely to produce outlier estimates, and “stretching” the curve honors that



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UVEAL MELANOMA and DEPRESSION

Uveal melanoma, a rare eye cancer, presents potential vision loss and life threat. This prospective, longitudinal study interrogated the predictive utility of visual impairment, as moderated by optimism/pessimism, on depressive symptoms in 299 adults undergoing diagnostic evaluation.



Annette
Stanton



James
MacDonald

MacDonald, J.J., Jorge-Miller, A., Enders, C.K., McCannel, T., Beran, T., & Stanton, A.L. (2021). Perceived and objective visual impairment predicting depressive symptoms across one year in uveal melanoma diagnostic biopsy: Optimism and pessimism as moderators. *Health Psychology, 40*, 408-417.

KEY VARIABLES



Depression

The CES-D is a 20-item inventory that asks people to rate how often they experience depressive symptoms such as restless sleep, poor appetite, and feeling lonely.



Cancer Diagnosis

Based on a diagnostic clinical evaluation for a possible intraocular malignancy, participants were classified as having malignant or nonmalignant diagnoses.

RESEARCH QUESTION

- Studies typically attempt to answer a handful of research questions involving associations between key variables
- Do people who receive a positive cancer diagnosis experience clinical levels of depressive symptoms, or do they show no meaningful elevation?
- CES-D scores > 16 are widely viewed as indicating risk for clinical depression, and scores < 16 are in the mild range

NULL HYPOTHESIS

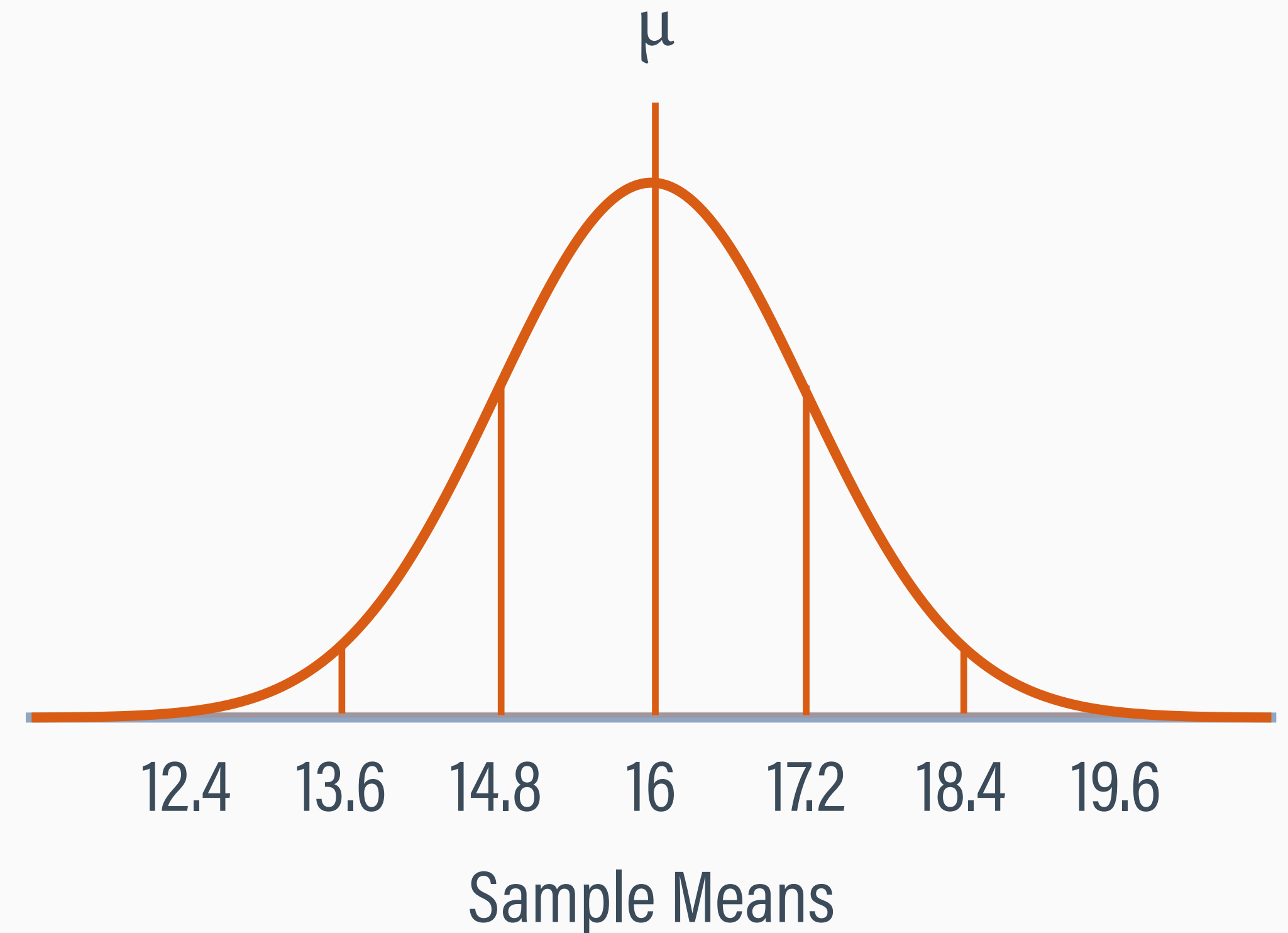
- The significance testing process starts with a null hypothesis that is counter to what we expect to find
- In the depression study, researchers want to determine whether or not people who receive a cancer diagnosis experience clinical depression
- The null that the true population mean is $\mu = 16$ is counter to expectations because researchers anticipate that the average level of depressive symptoms will exceed the clinical cutoff

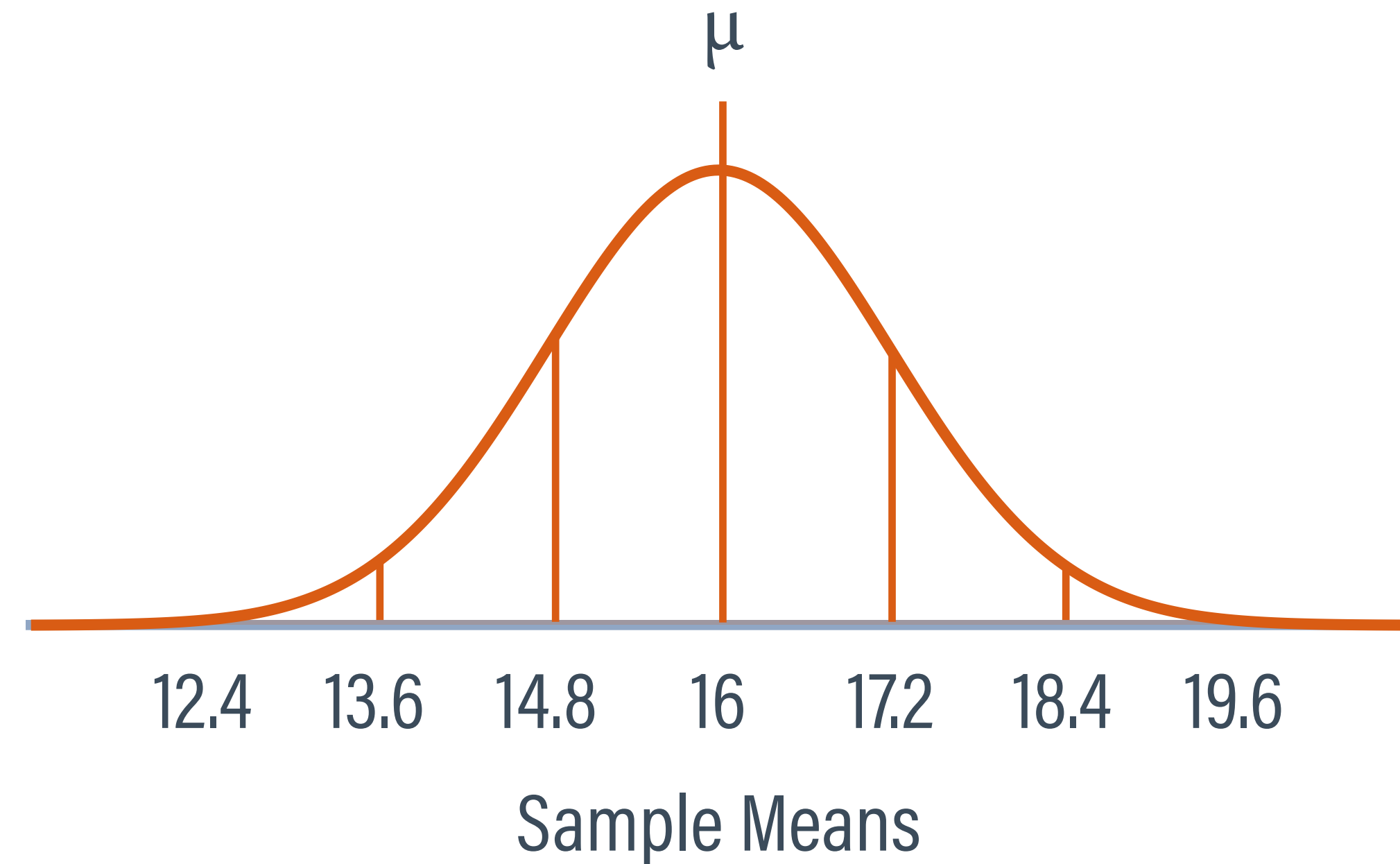
DEMONSTRATING AN EFFECT

- The null hypothesis that is counter to what we expect to find
- To demonstrate an effect (e.g., patients experience clinical symptoms), we attempt to refute the null with evidence from the data
- To refute the null that $\mu = 16$, we would need to observe a sample mean $\bar{X}$ that is much higher or much lower than 16

SAMPLING DISTRIBUTION IF NULL IS TRUE

- The sampling distribution shows the means from many hypothetical samples of $N = 107$ drawn out of the null population with $\mu = 16$
- To refute the null, the sample $\bar{X}$ must lie far away from 16 (either above or below)
- An $\bar{X}$ close to 16 would lend support to the null

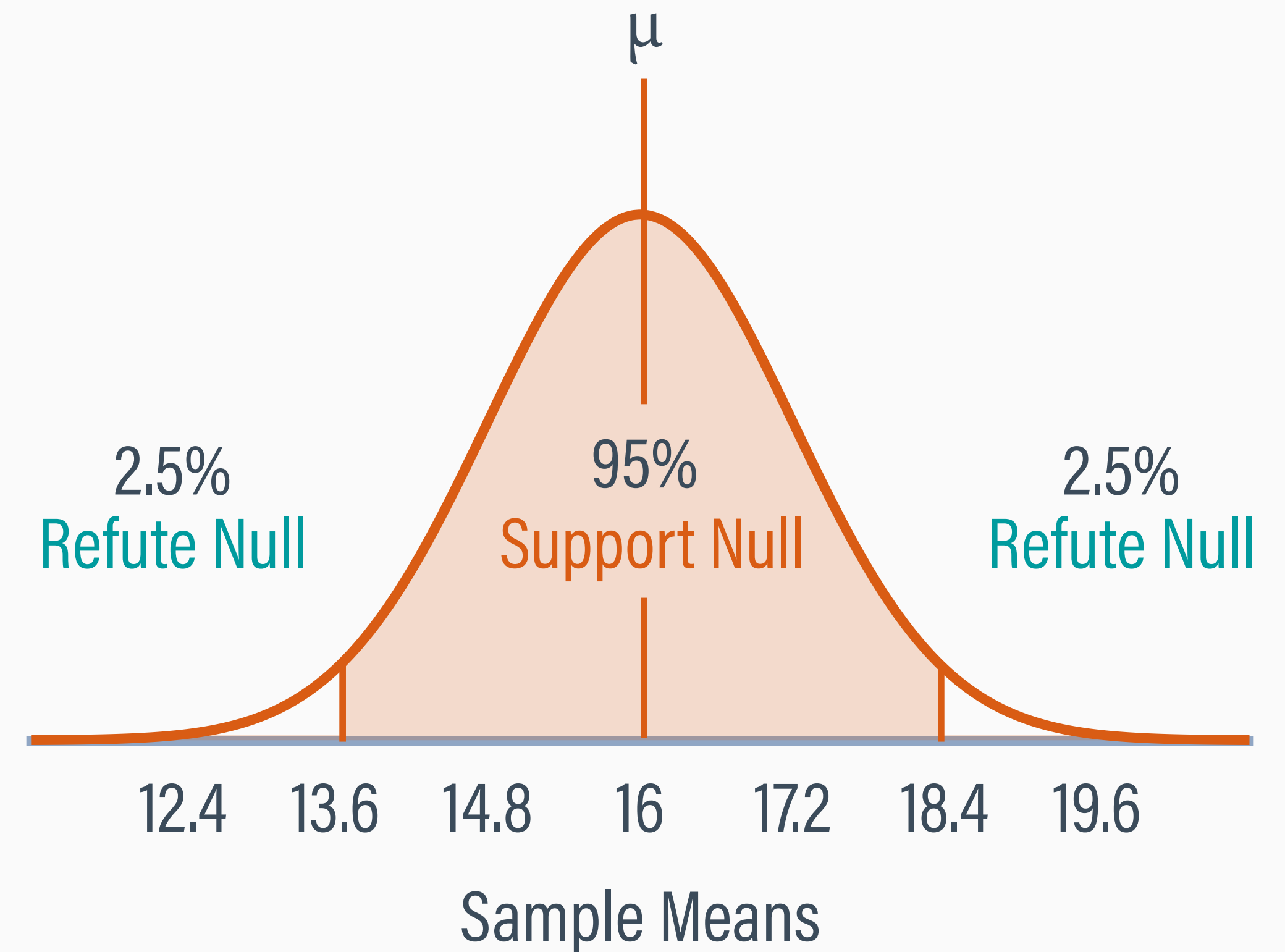




Consider the sampling distribution of sample means from a null population with $\mu = 16$. In small groups of two or three, discuss how far above or below 16 the sample mean ($\bar{X}$) would need to be for you to feel confident that the data refutes the null (i.e., demonstrates an effect).

EVALUATING THE NULL

- By convention, we refute the null if the sample $\bar{X}$ falls outside the middle 95% of the sampling distribution
- Such as sample has less than a 5% chance of originating from the null population
- Any $\bar{X}$ within the middle 95% of the sampling distribution lends support to the null



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NULL HYPOTHESIS

- In the population, people who receive a cancer diagnosis have a mean exactly at the clinical cutoff 16

$$H_0: \mu = 16$$

- The null that $\mu = 16$ is counter to expectations because researchers anticipate that depressive symptoms could exceed (or possibly fall below) the clinical cutoff

ALTERNATE HYPOTHESIS

- The alternate hypothesis reflects the researcher's expectations about an effect
- A one-tailed (directional) alternate hypothesis predicts an effect in just one direction (e.g., people are above the clinical cutoff), and a two-tailed alternate predicts an effect in either direction (e.g., people could be above or below the clinical cutoff)
- Two-tailed tests predominate psychology applications

TWO POSSIBLE ALTERNATIVE HYPOTHESES

- One-tailed alternate: People who receive a cancer diagnosis are above the clinical cutoff for depression

$$H_A: \mu > 16$$

- Two-tailed alternate: People who receive a cancer diagnosis could be above or below the clinical cutoff for depression

$$H_A: \mu \neq 16 (\mu > 16 \text{ or } \mu < 16)$$

ROLE OF THE ALTERNATE HYPOTHESIS

- We do not directly test the alternative hypothesis, which represents the researcher's belief
- We provide support for the alternative by rejecting the null
- Whether the alternative is one-tailed or two-tailed is very important, as this determines what kind of evidence would lead us to reject the null

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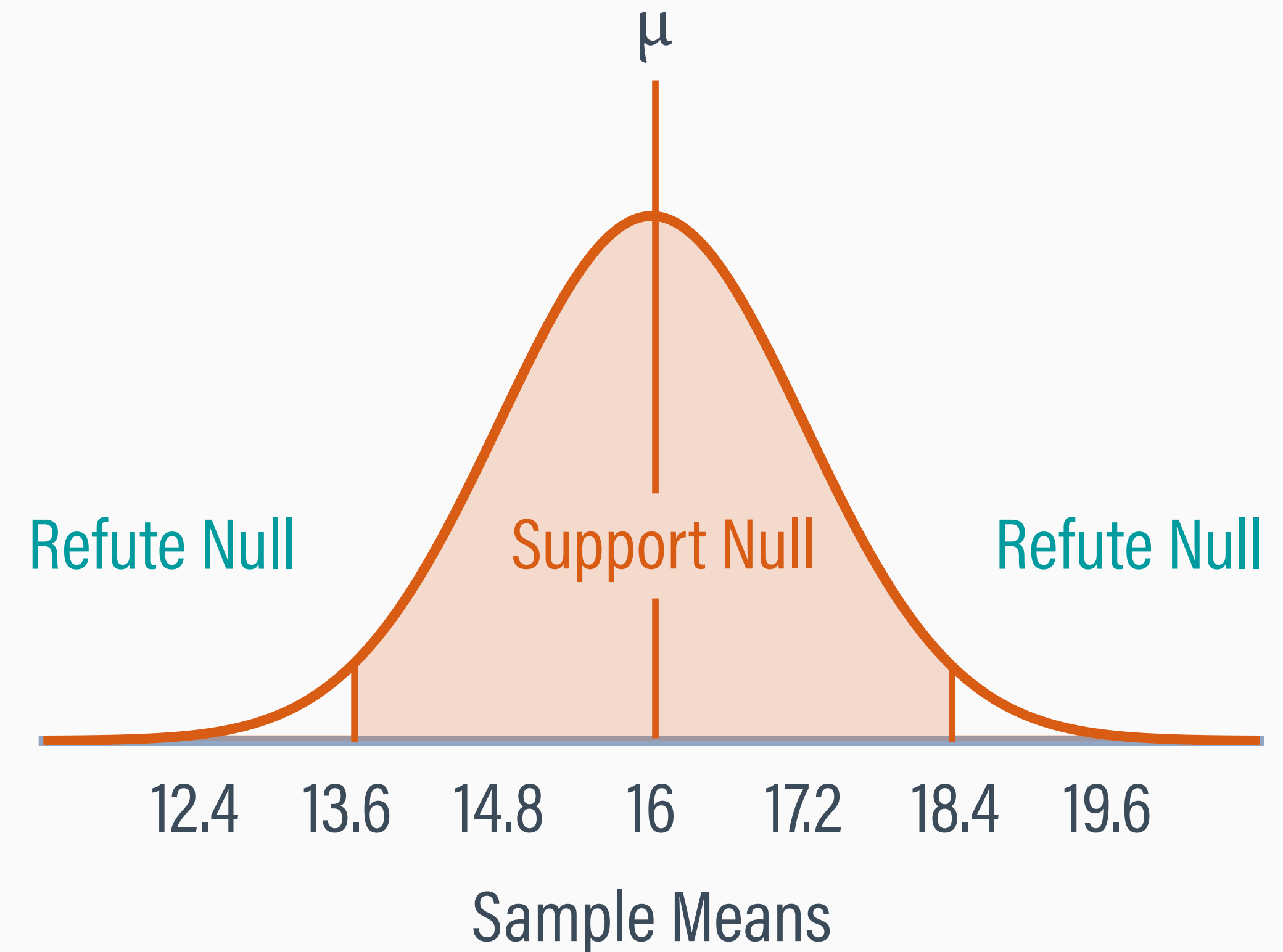
Evaluate hypotheses and draw conclusion

STANDARD OF EVIDENCE

- The data are the evidence that we use to conclude whether the null is plausible (“innocent”) or implausible (“guilty”)
- If the sample mean from our data is very different from the null mean, then we conclude that the null hypothesis is implausible
- How big a difference do we need to observe to refute the null?

EVALUATING THE NULL

- Any $\bar{X}$ near the middle of the sampling distribution lends support to the null
- Such as sample has a high probability of originating from the null population
- We refute the null if the sample $\bar{X}$ falls far from μ
- Such as sample has a low probability of originating from the null population



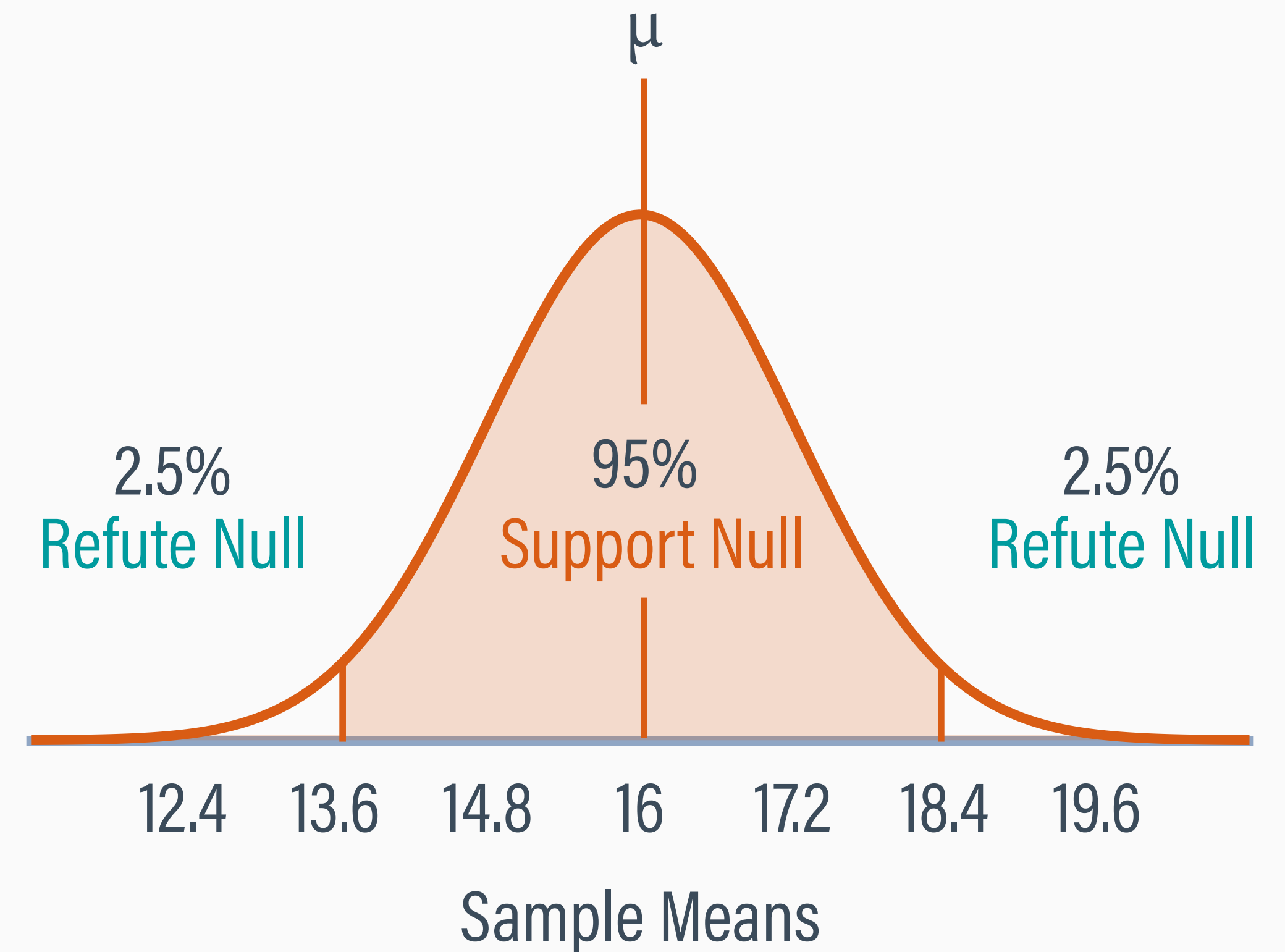
5% SIGNIFICANCE CRITERION

- Evidence against the null is presented as a probability
- R.A. Fisher — an influential biologist and statistician from the early 1900s — recommended 5% ($p < .05$) as a criterion
- If a sample mean has a 5% or less probability of originating from the null population, then we conclude H_0 is implausible



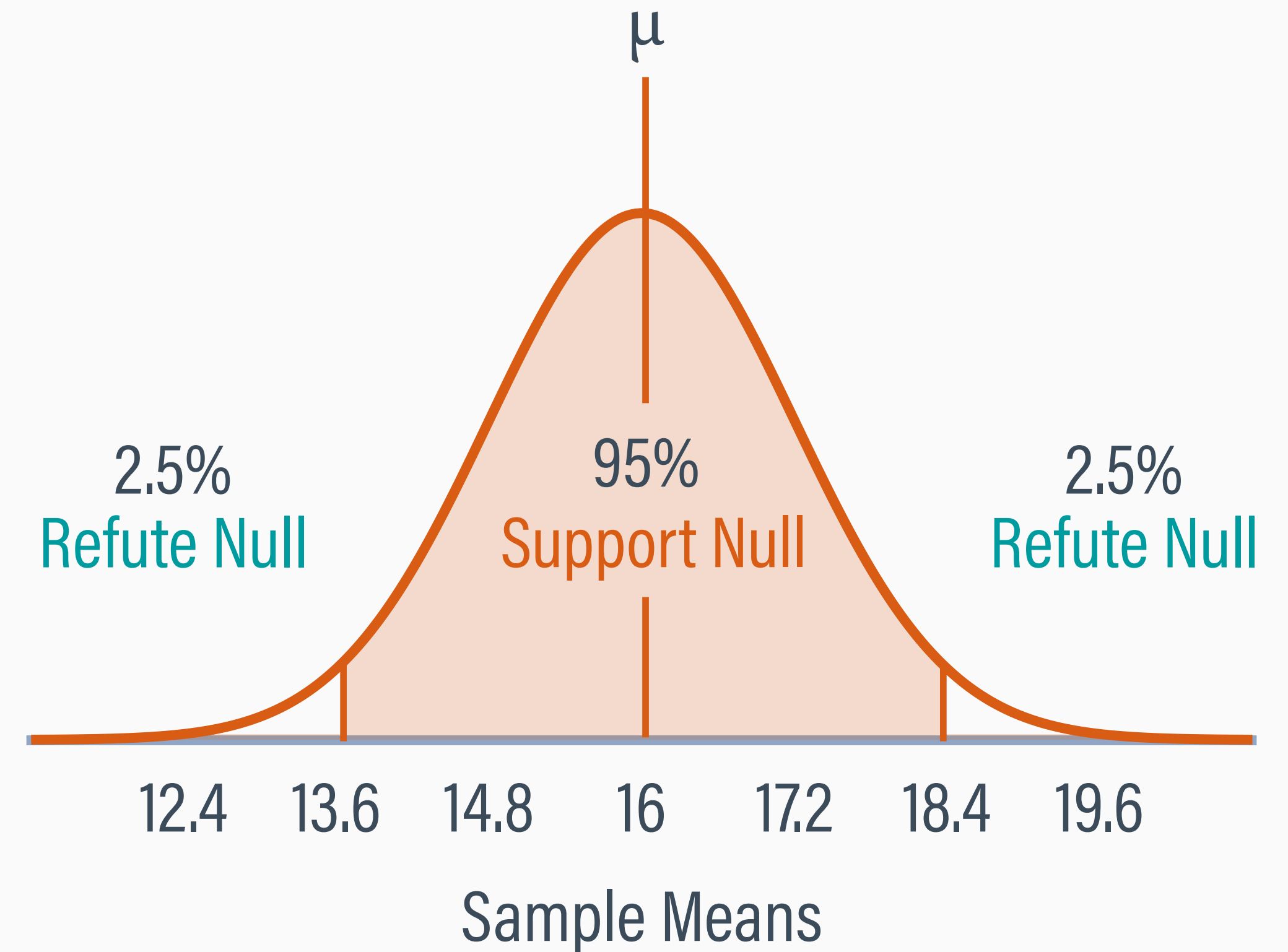
SAMPLING DISTRIBUTION

- By convention, we refute the null if the sample $\bar{X}$ falls outside the middle 95% of the sampling distribution
- Such as sample has less than a 5% chance of originating from the null population
- That is, we refute the null if the sample $\bar{X}$ is an outlier (rare) from the null population ($p < .05$)



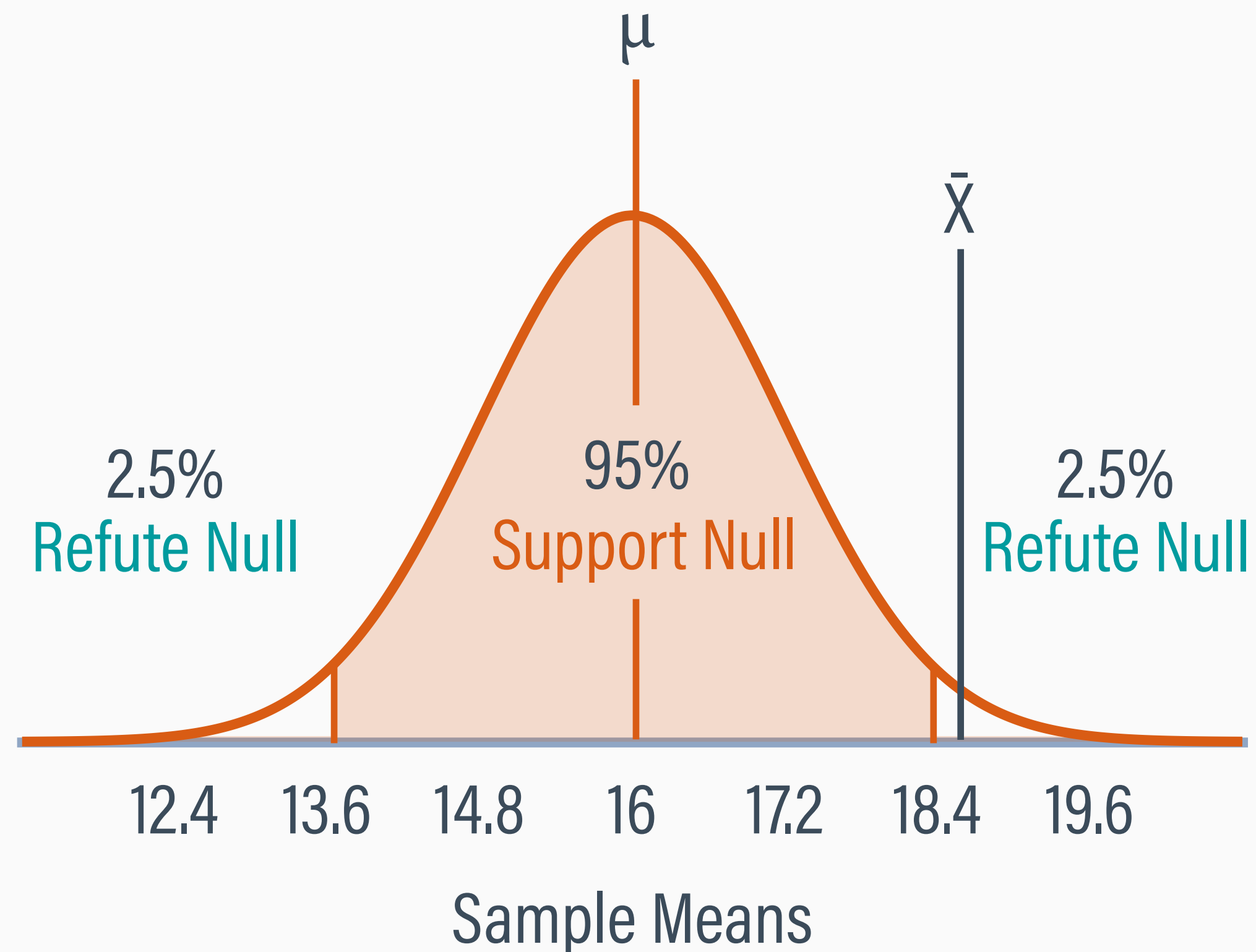
TWO-TAILED ALTERNATE HYPOTHESES

- A two-tailed alternate hypothesis predicts an effect in either direction
- e.g., People who receive a cancer diagnosis could be above or below the clinical cutoff
- The 5% rejection region (**alpha level**) is split in half to allow for the possibility that either an increase or a decrease provides evidence against H_0

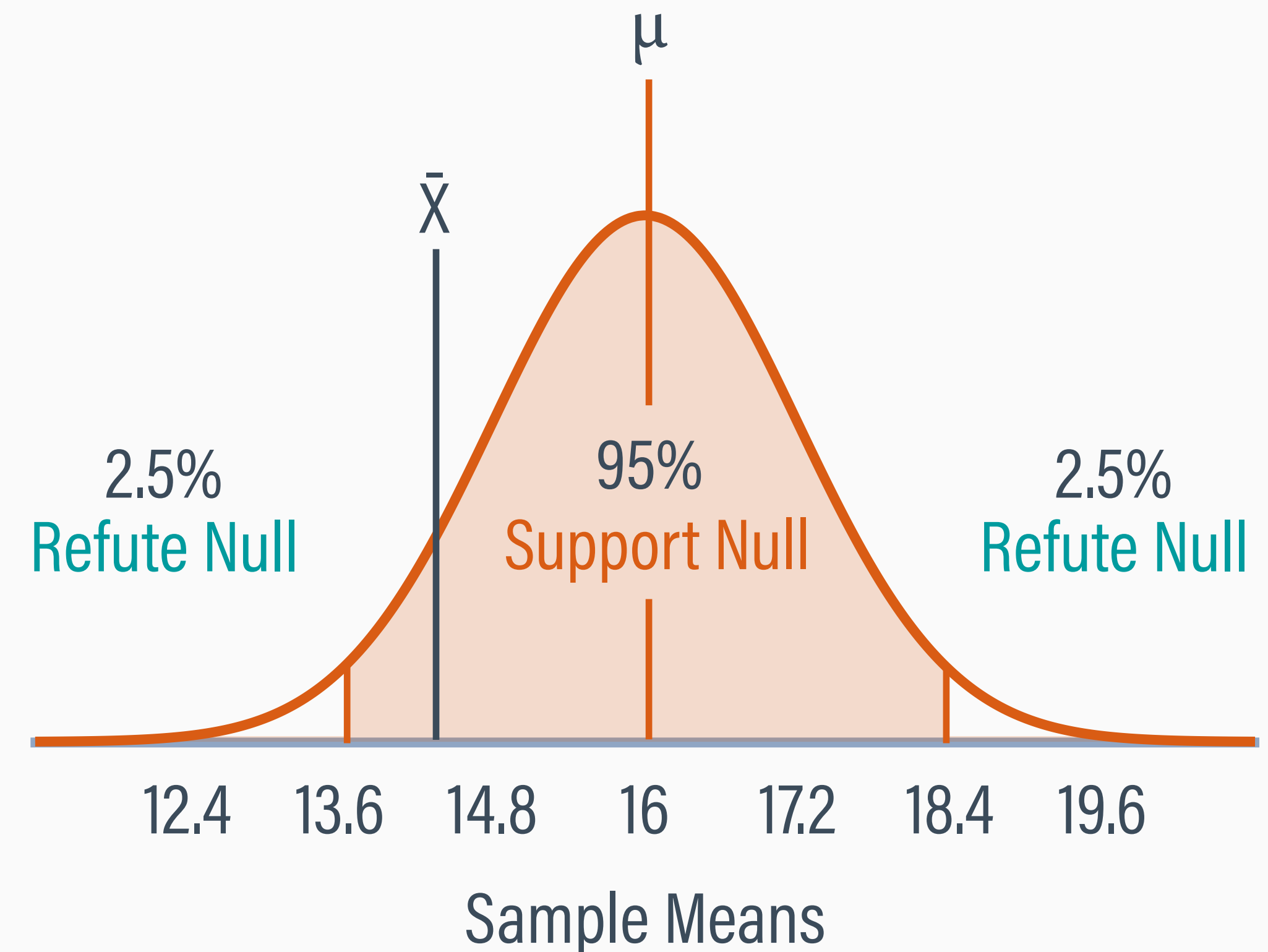


TWO POSSIBLE DECISIONS

Refute the null: $\bar{X}$ is unlikely from a population with $\mu = 16$ ($p < .05$)

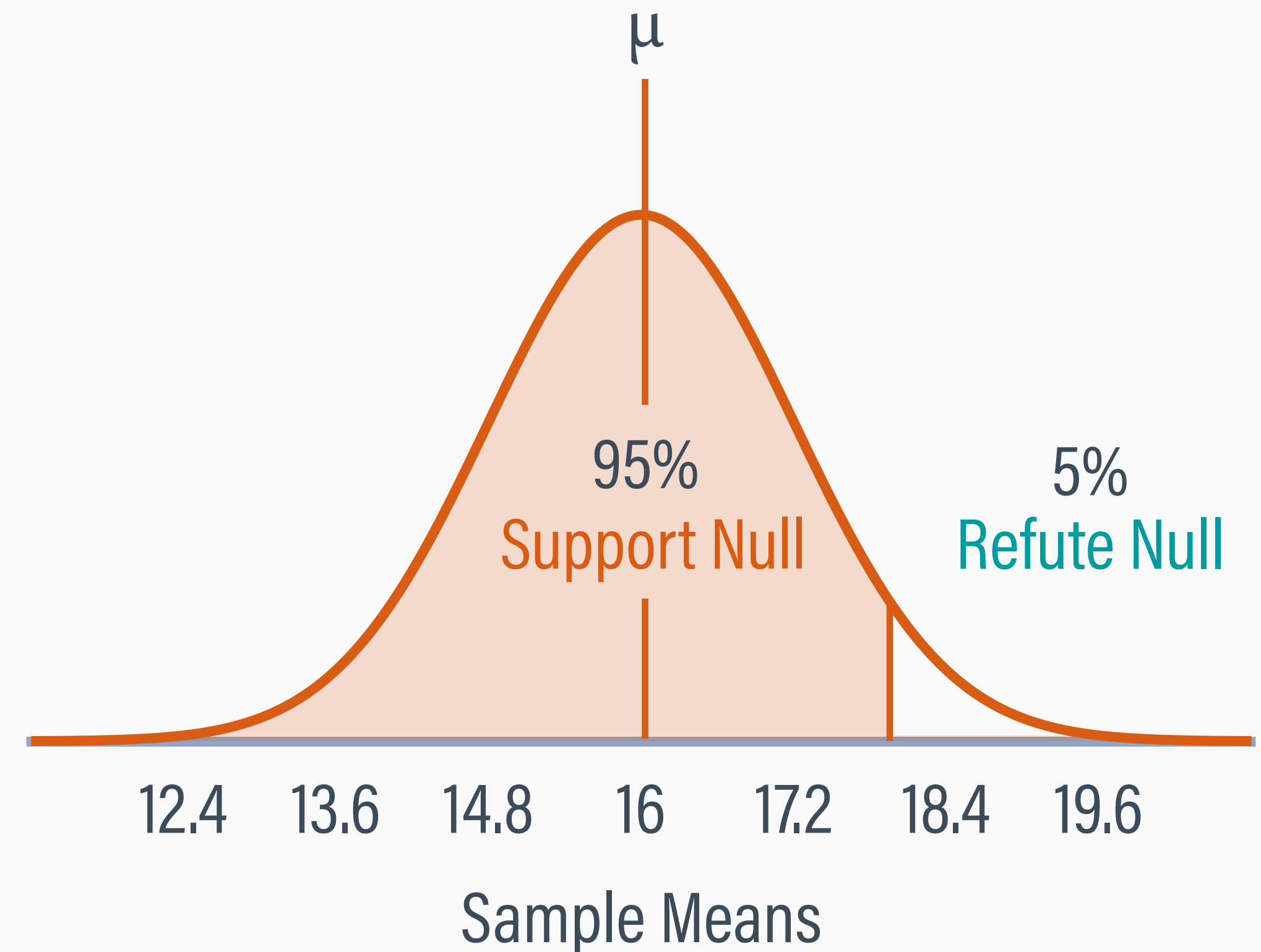


Support the null: $\bar{X}$ is likely from a population with $\mu = 16$ ($p > .05$)



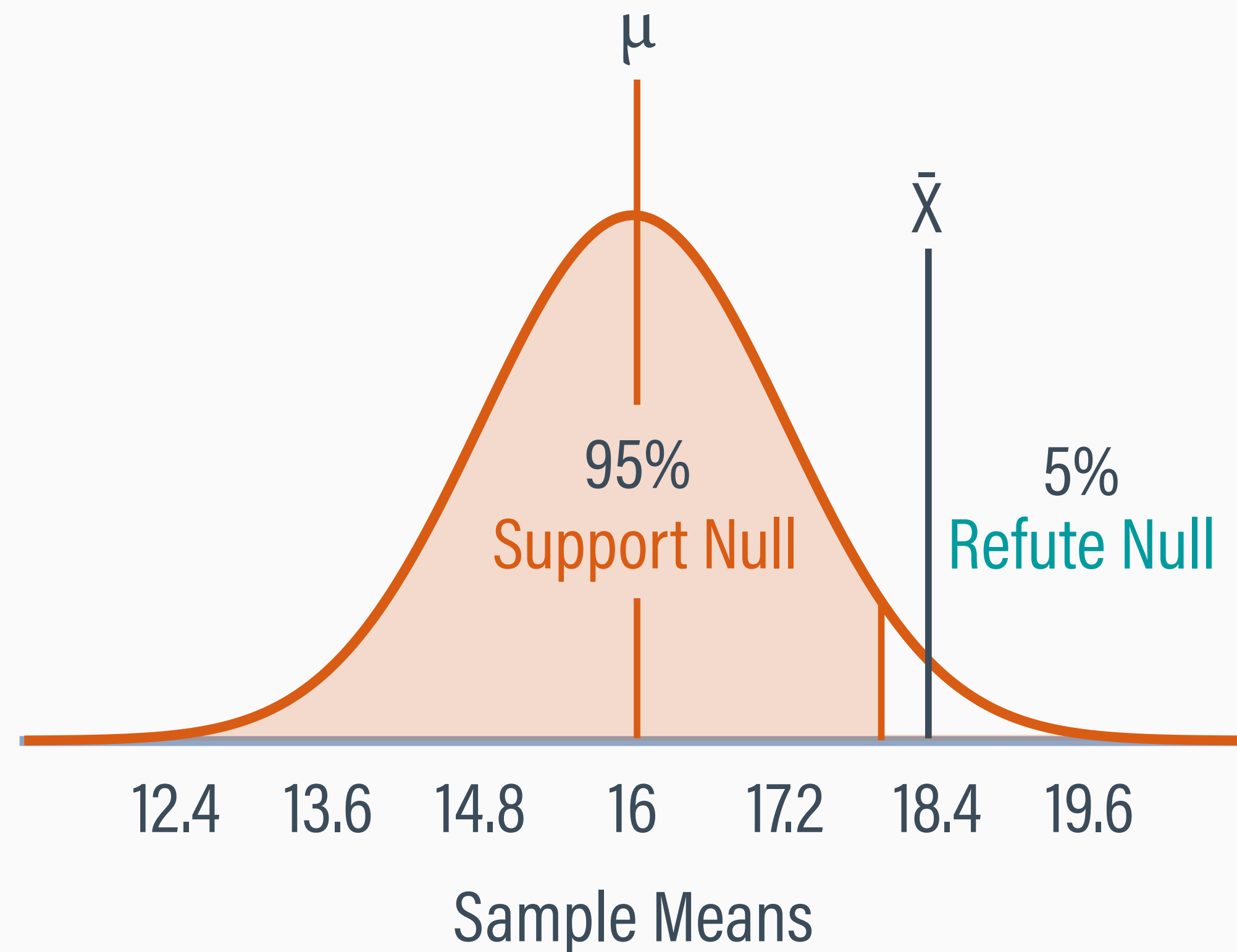
ONE-TAILED ALTERNATE HYPOTHESES

- A one-tailed alternate hypothesis predicts an effect in just one direction
- e.g., People who receive a cancer diagnosis could only be above the clinical cutoff
- The 5% rejection region (**alpha level**) is assigned to one tail to reflect that only a very high $\bar{X}$ can provide evidence against H_0

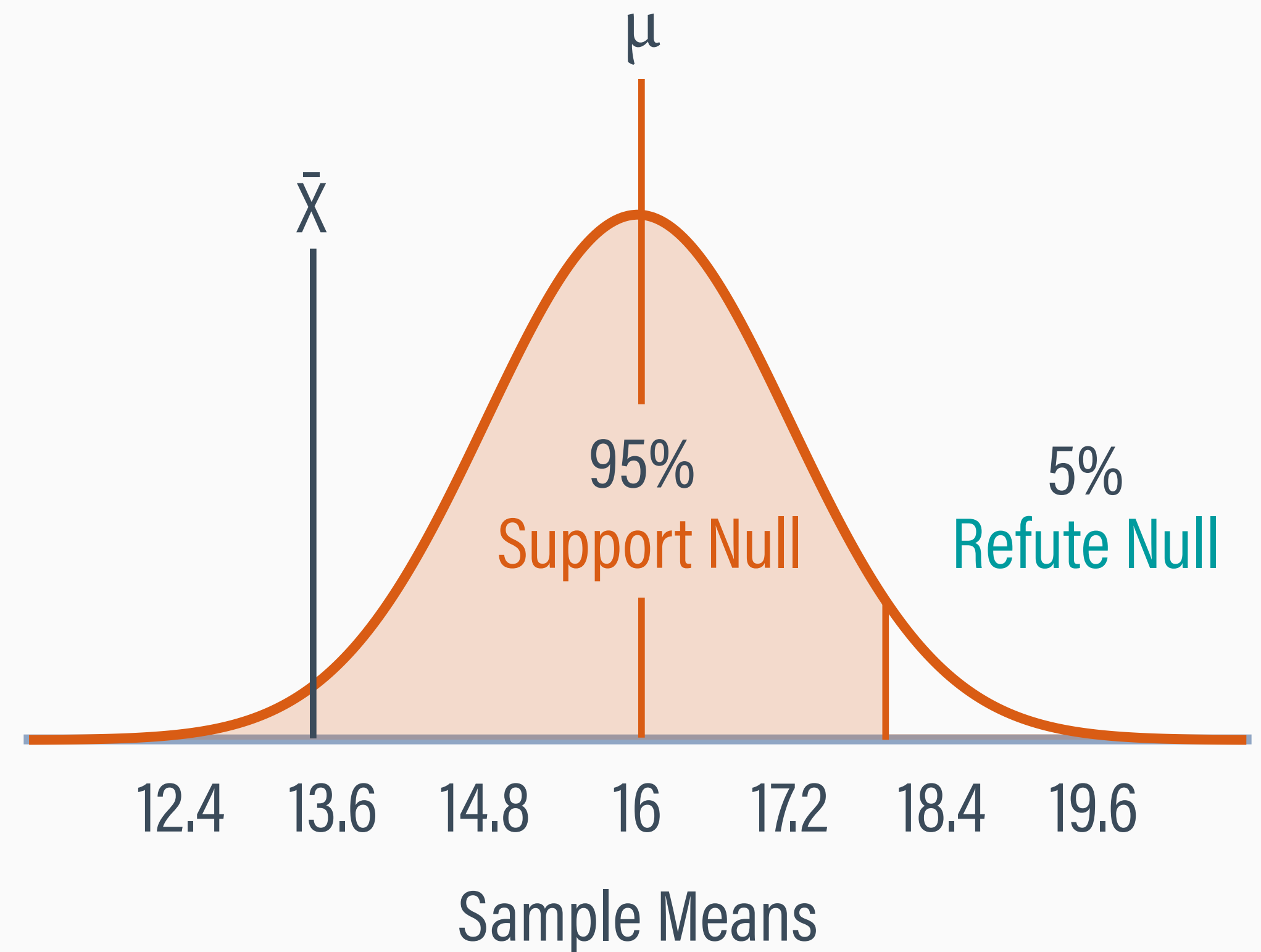


TWO POSSIBLE DECISIONS

Refute the null: $\bar{X}$ is unlikely from a population with $\mu = 16$ ($p < .05$)



Support the null: $\bar{X}$ is likely from a population with $\mu = 16$ ($p > .05$)



FALSE POSITIVES (TYPE I ERRORS)

- The 5% criterion defines a region of the distribution that contains outlier samples that are unlikely *but not impossible*
- When $\bar{X}$ falls in the rejection region (evidence against the null), there is still a 5% chance it came from the null population
- A significant result means we risk a false positive—incorrectly rejecting the null when it is actually true (called a **Type I error**)
- Adopting a 5% criterion fixes the Type I error risk to 5%

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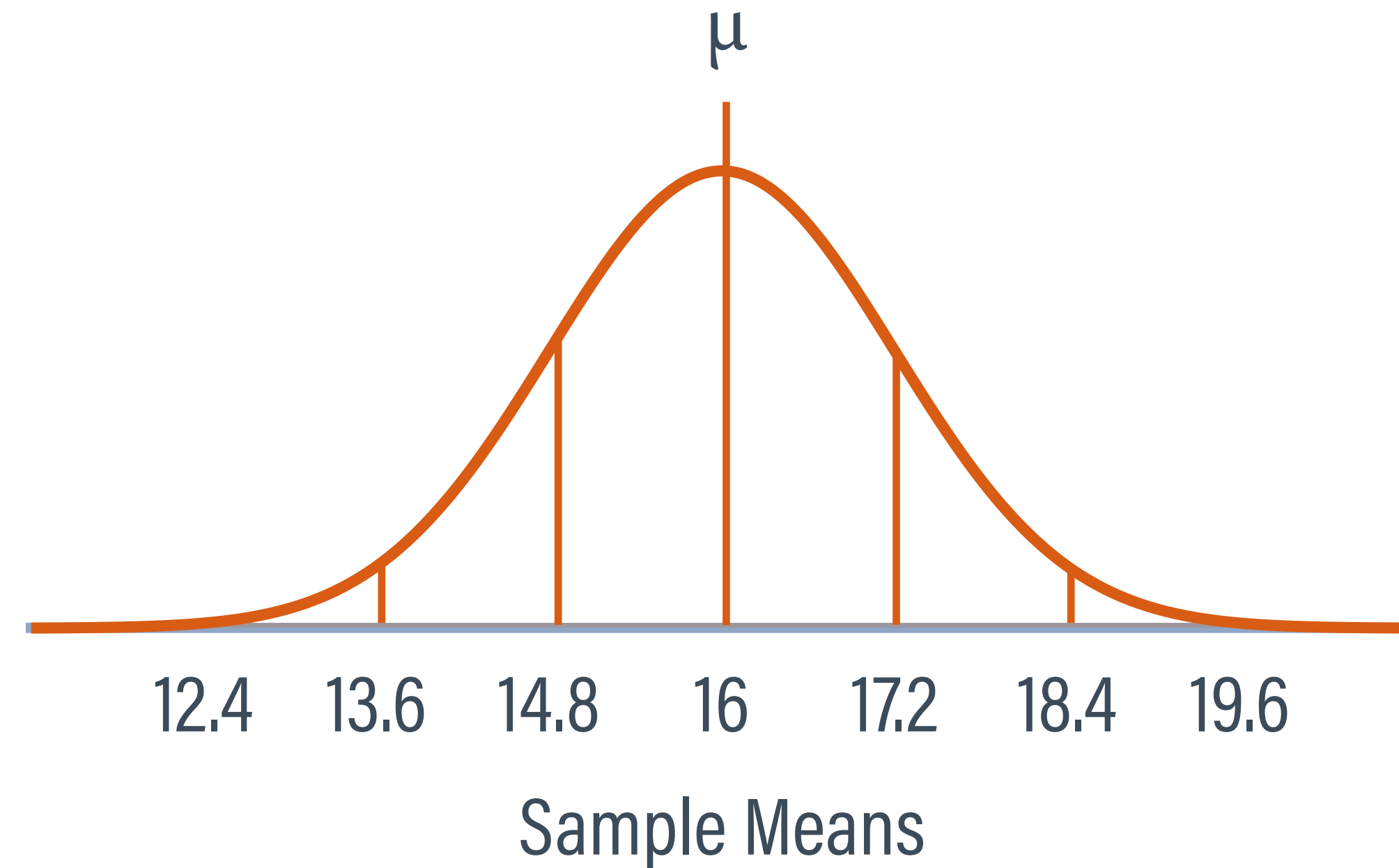
Evaluate hypotheses and draw conclusion

ANALYSIS SUMMARY

- N = 107 participants received a cancer diagnosis
- The sample mean is 18.10
- The standard error $s_{\bar{X}} = 1.20$ indicates that the expected (average) sampling error across many samples of N = 107 is about 1.20 CES-D points

N	$\bar{X}$	SD	SE
107	18.10	12.45	1.20

$$s_{\bar{X}} = \frac{s}{\sqrt{N}} = \frac{12.45}{\sqrt{107}} = 1.20$$



Consider the sampling distribution of sample means from a null population with $\mu = 16$. In small groups of two or three, discuss the descriptive statistics and determine whether $\bar{X} = 18.10$ ($s_{\bar{X}} = 1.20$) provides evidence for or against the null hypothesis.

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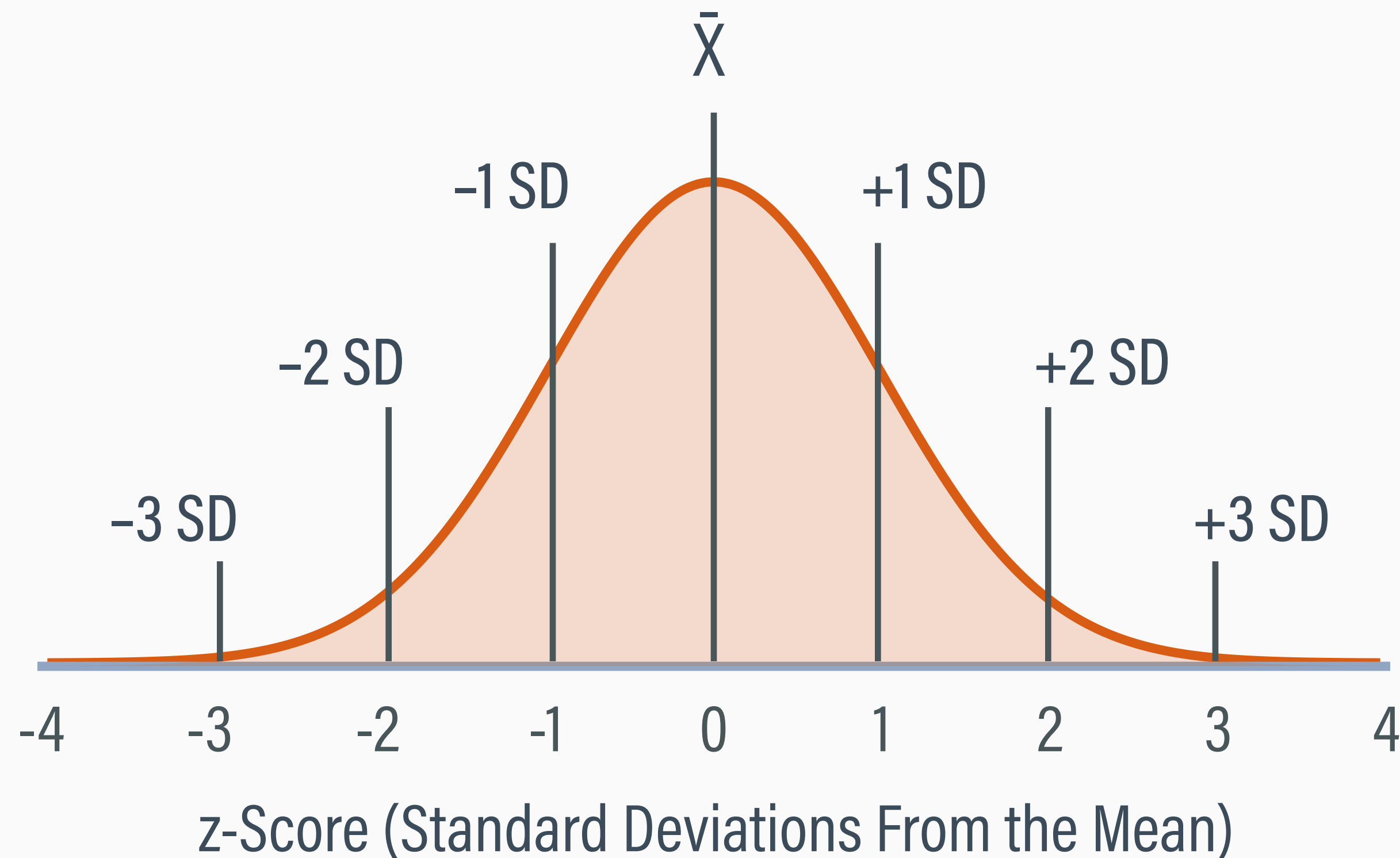
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COMPARING DATA TO THE NULL

- Two ways to determine whether the sample $\bar{X}$ is consistent (or inconsistent) with the null population mean
- The t-statistic gives a standardized distance between the sample mean and the null hypothesis mean (like a z-score)
- A p-value tells us how likely it is that hypothetical samples like our data would originate from the null population

Z-SCORES REVISITED

- The z-score scale is a common standardized metric that expresses scores as standard deviation units from the mean



Z-SCORE FORMULA

- ◉ The numerator captures the raw score's distance (deviation) from the sample mean $\bar{X}$

$$z = \frac{X - \bar{X}}{s} = \frac{\text{distance from mean}}{\text{standard deviation}}$$

- ◉ Dividing by the standard deviation converts the distance to standard deviation units

t-STATISTIC

- The t-statistic quantifies the number of standard error units that separate the sample mean and null hypothesis population mean

$$t = \frac{\bar{X} - \mu}{s_{\bar{X}}} = \frac{\text{distance from the null}}{\text{standard error (std. dev. of } \bar{X})}$$

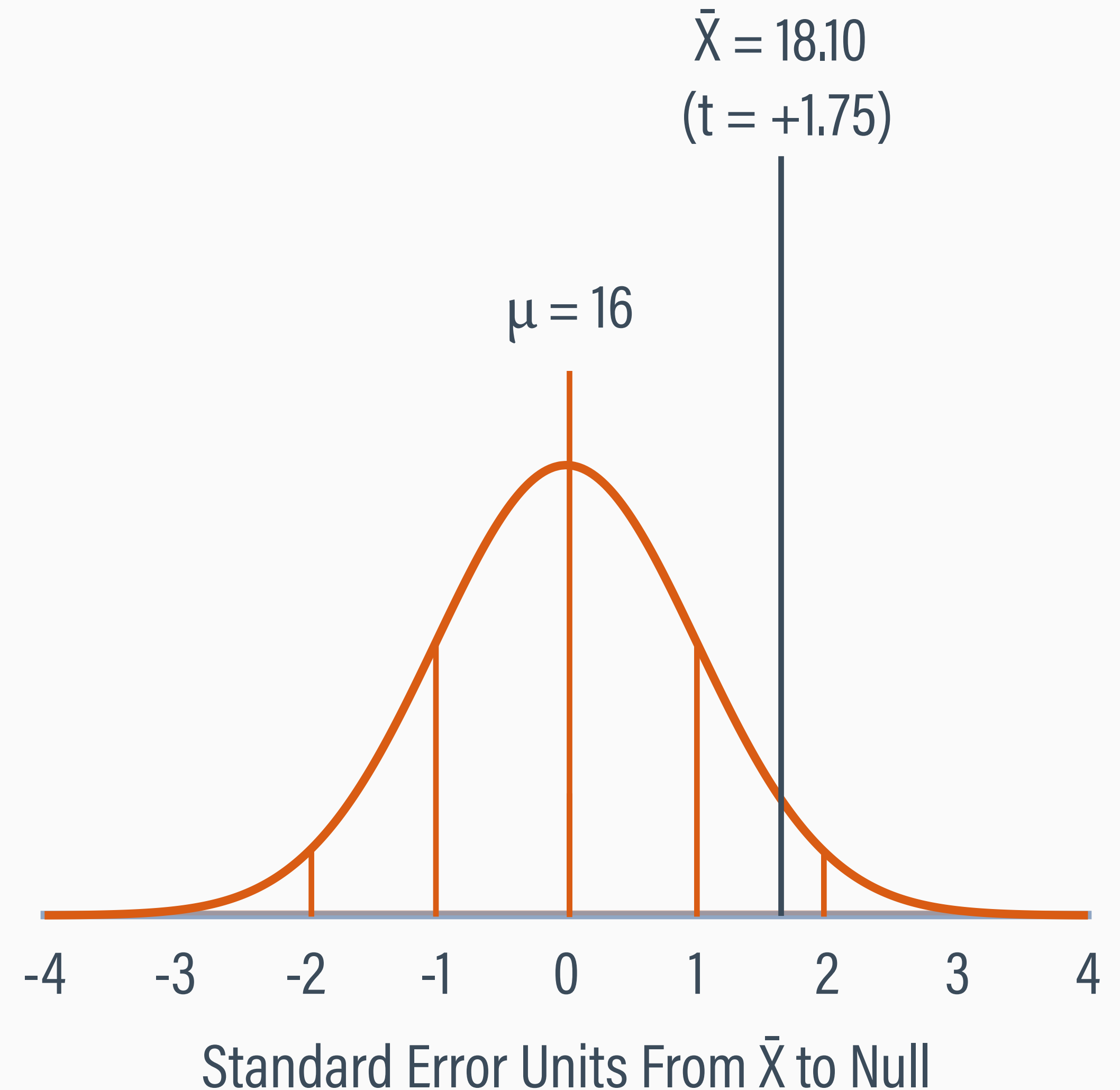
- The t-statistic is the same as a z-score (a standardized metric where distance is expressed in standard deviation units)

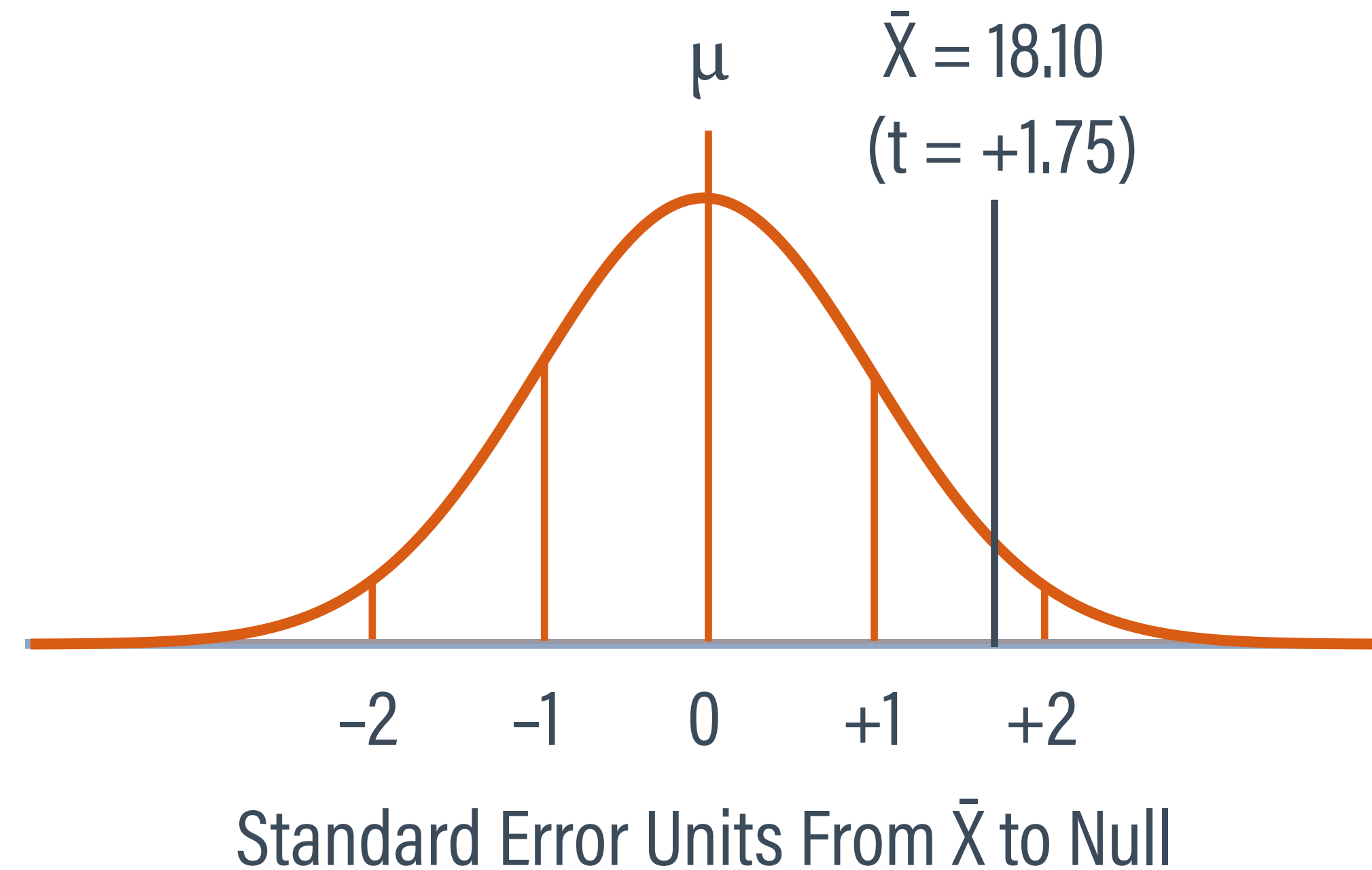
t-STATISTIC EXAMPLE

- The positive t-statistic conveys that the sample $\bar{X}$ is higher than the null ($\mu = 16$)

$$t = \frac{\bar{X} - \mu}{s_{\bar{X}}} = \frac{18.10 - 16}{1.20} = 1.75$$

- The numeric value indicates that $\bar{X}$ is 1.75 standard error units from the null null





Consider the sampling distribution of sample means from a null population with $\mu = 16$. The sample mean and t-statistic are $\bar{X} = 18.10$ and $t = 1.75$. In small groups of two or three, apply the normal curve rule of thumb and decide whether the sample data provide evidence for or against the null hypothesis

FREQUENTIST PARADIGM REVISITED

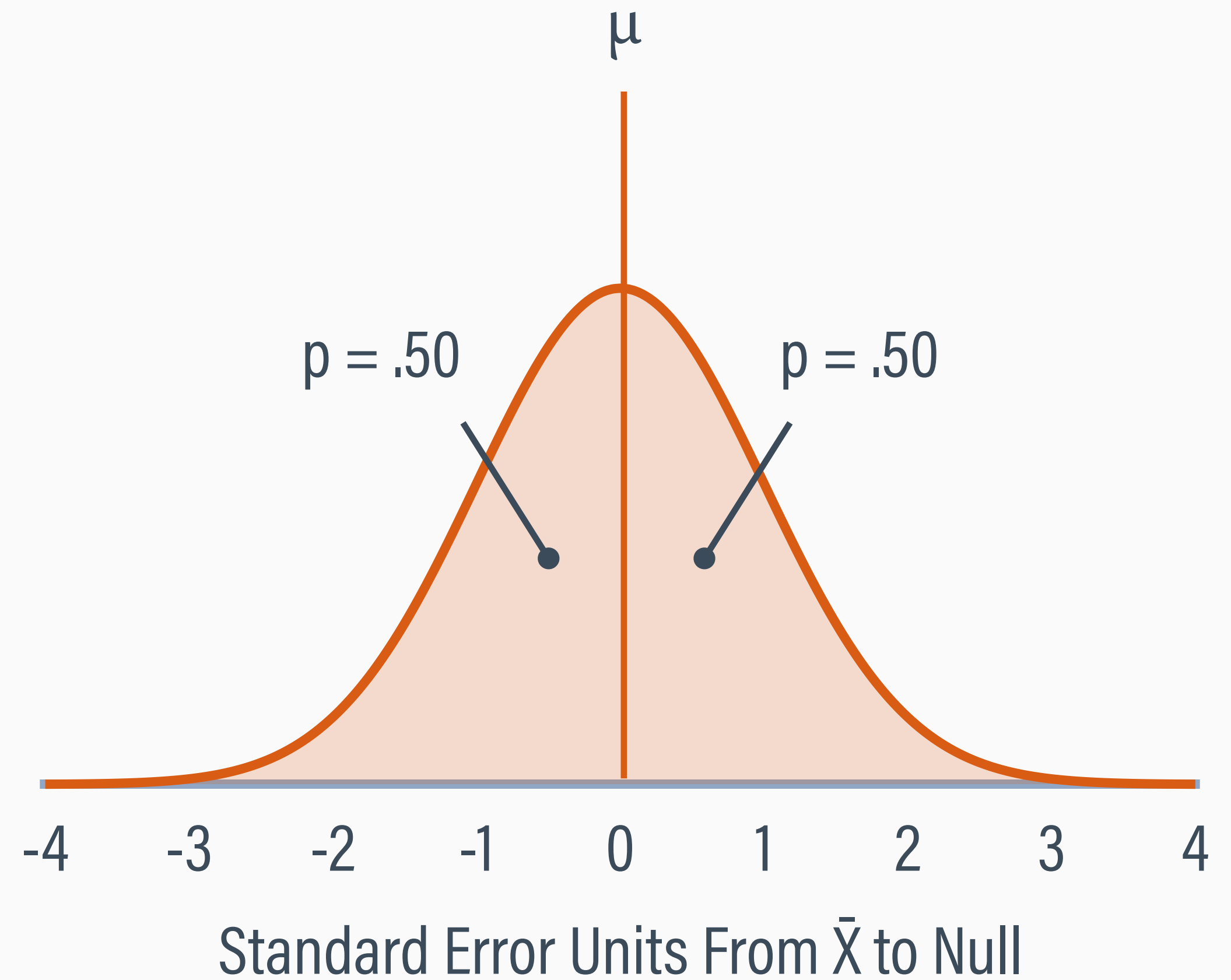
- The frequentist paradigm is defined by the idea that there is one population with unknown parameters (e.g., μ and σ)
- We imagine numerous hypothetical samples of size N from the population, each with its own unique estimates and 95% CIs
- The population-level statistics (parameters) are locked in at a single set of values, whereas the sample-level statistics (estimates, t-statistics) vary across different data sets

PROBABILITY VALUES (P-VALUES)

- A p-value is defined as proportion of hypothetical samples that have a t-statistic at least as large as the sample data
- Assuming the null is true, how likely is it to draw a sample with an effect at least as large as the one from our data?
- Visually, probability is an area under the curve, obtained by applying calculus integrals to the t-distribution function

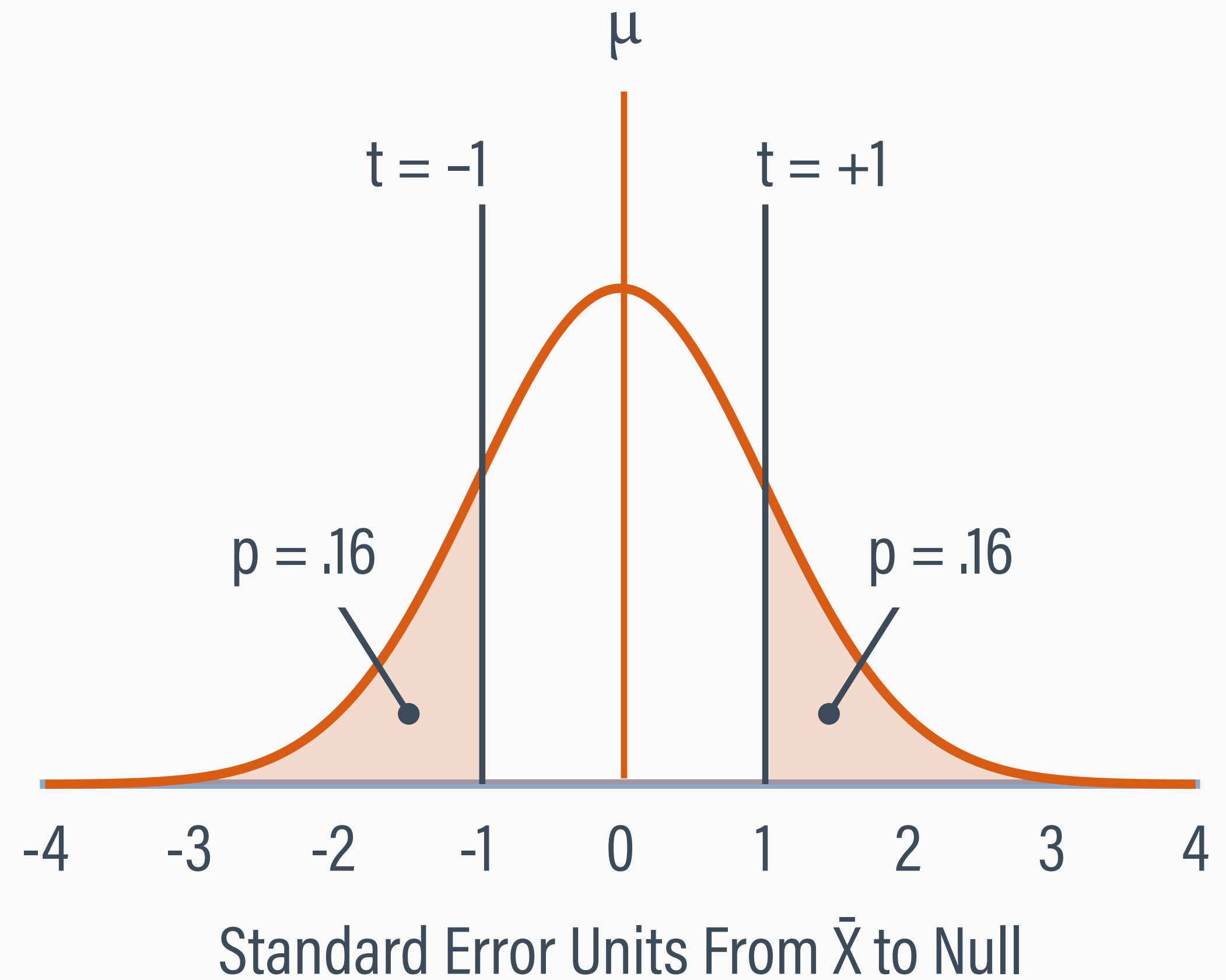
TOTAL AREA UNDER THE CURVE

- The total probability (area under the sampling distribution) in both directions is 1.0
- 50% of all hypothetical samples are above the null population mean and 50% are below



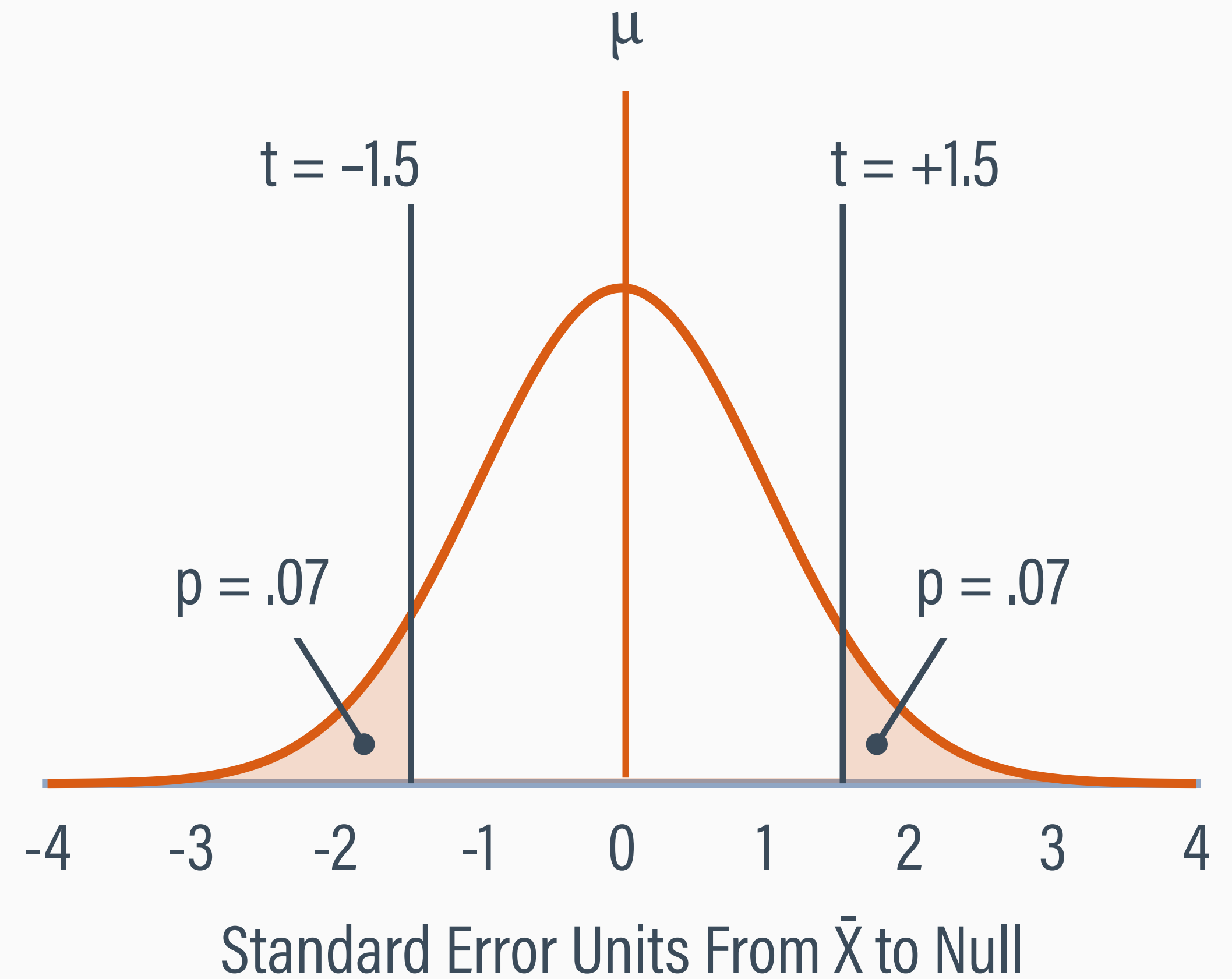
P-VALUE AT ± 1 STANDARD ERRORS

- 16% of all hypothetical samples are +1 or more standard error units above the null mean, and 16% are -1 or more standard errors below
- The two-tailed probability of drawing a sample from the null population with a t-statistic of at least ± 1 is .32 (.16 positive + .16 negative)



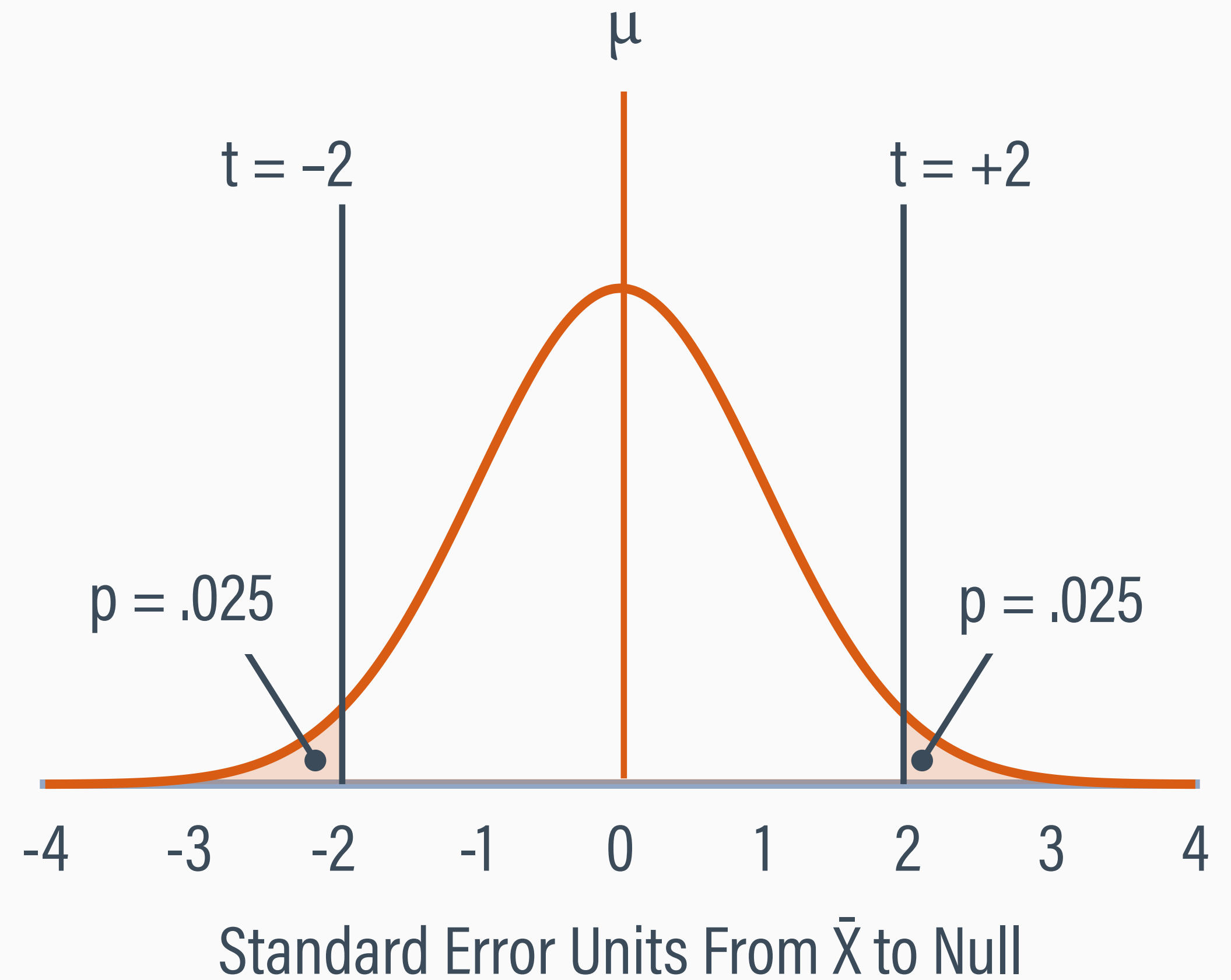
P-VALUE AT ± 1.5 STANDARD ERRORS

- 7% of all hypothetical samples are +1.5 or more standard error units above the null mean, and 7% are -1.5 or more standard errors below
- The two-tailed probability of drawing a sample from the null population with a t-statistic of at least ± 1 is .14 (.07 positive + .07 negative)



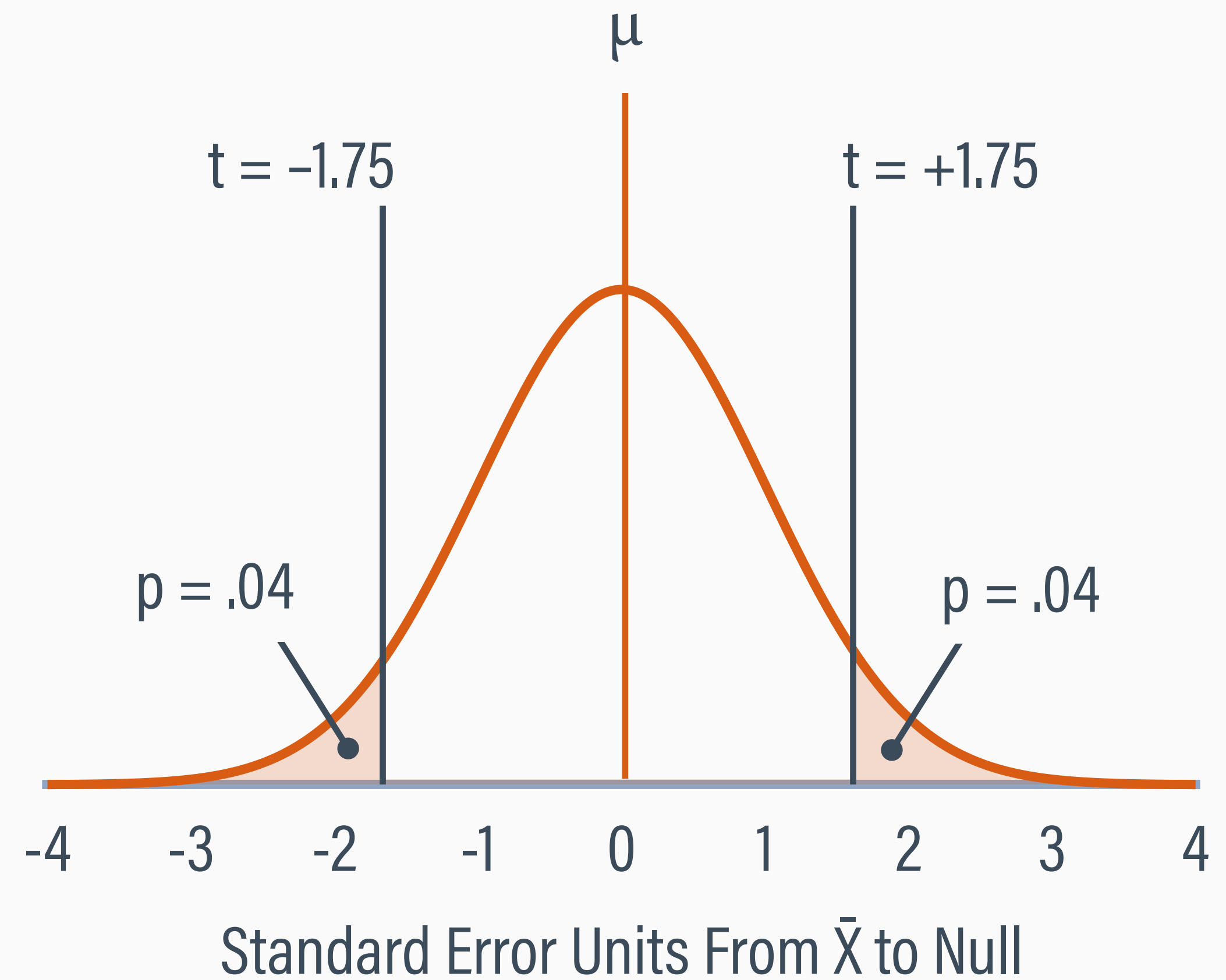
P-VALUE AT ± 2 STANDARD ERRORS

- 2.5% of all hypothetical samples are at least +1.96 standard error units above the null mean, and 2.5% are -1.96 or more below
- The two-tailed probability of drawing a sample from the null population with a t-statistic of at least ± 1.96 is .05 (.025 positive + .025 negative)



TWO-TAILED P-VALUE FOR DEPRESSION DATA

- Depress mean: $\bar{X} = 18.10$, $t = 1.75$
- The two-tailed probability of drawing a sample from the null population with a t-statistic of at least ± 1.75 is $p = .08$
- The probability of drawing a sample mean at least as extreme as ours from a null population with $\mu = 16$ is about 8%

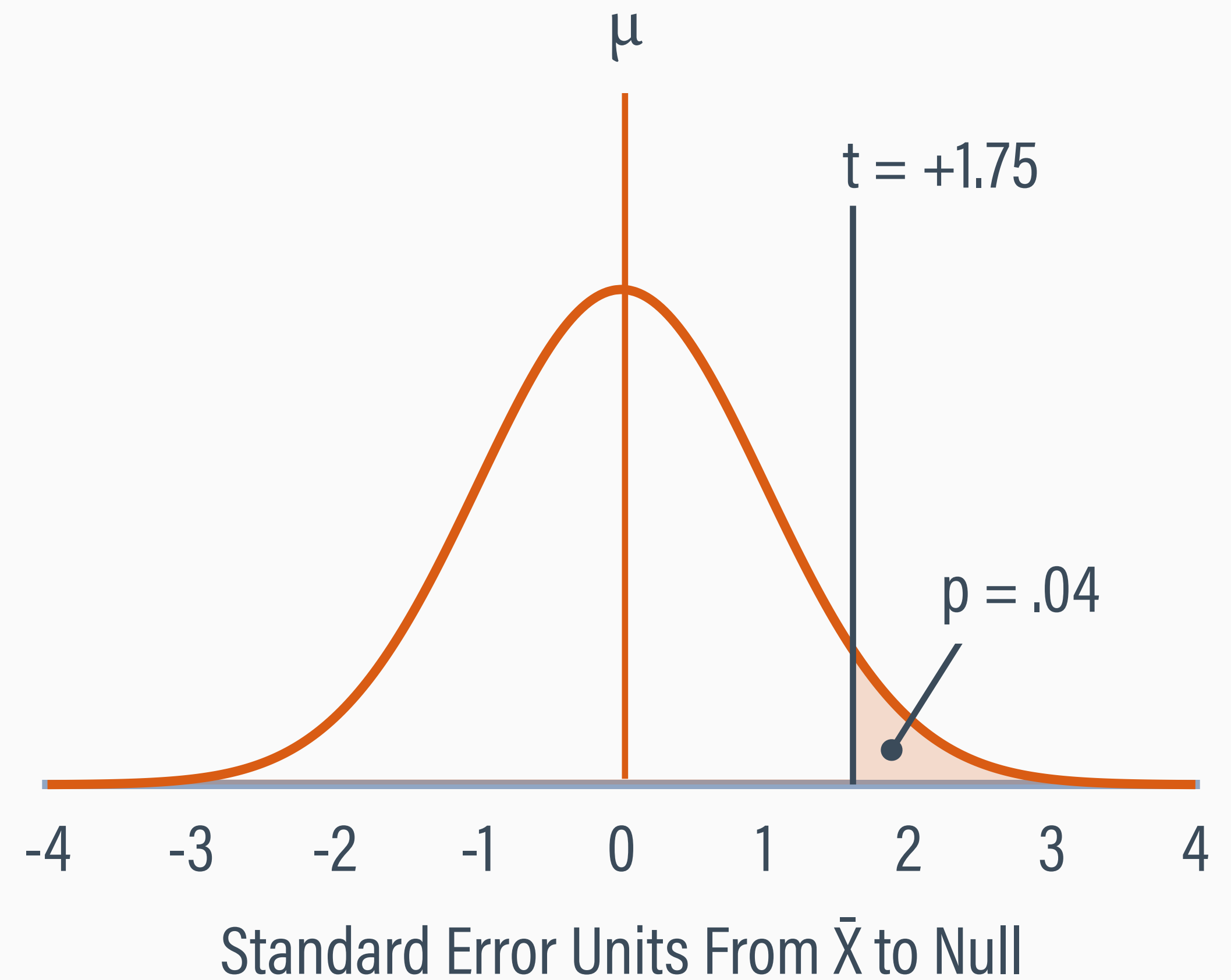




Consider the following statement, which is a common misinterpretation about the probability value: $p = 0.08$ means there's a 8% chance the null is true (and 92% chance the alternative is true). In small groups of two or three, discuss why this interpretation is incorrect. Hint: Think about the frequentist statistical paradigm.

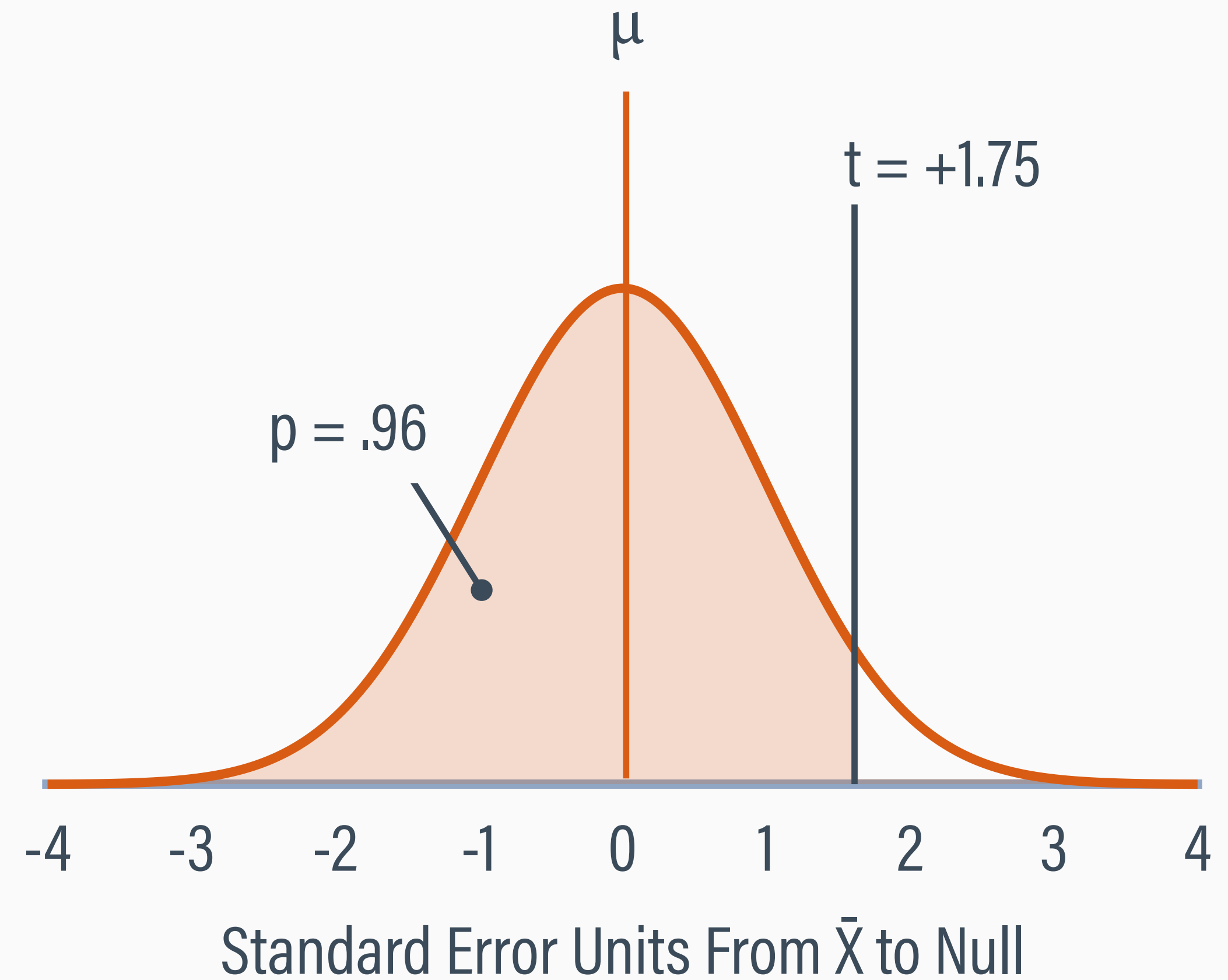
ONE-TAILED P-VALUE FOR DEPRESSION DATA

- Depress mean: $\bar{X} = 18.10$, $t = 1.75$
- Suppose the researchers specified a one-tailed hypothesis where only an increase in depression could refute the null
- The one-tailed probability of drawing a sample from the null population with a t-statistic of at least +1.75 is $p = .04$



ONE-TAILED P-VALUE FOR DEPRESSION DATA

- Depress mean: $\bar{X} = 18.10$, $t = 1.75$
- Suppose the researchers specified a one-tailed hypothesis where only an decrease in depression could refute the null
- The sample mean is in the wrong direction!
- The one-tailed probability of drawing a sample from the null population with a t-statistic *lower than* +1.75 (in the hypothesized direction) is $p = .96$



SIGNIFICANCE TESTING STEPS

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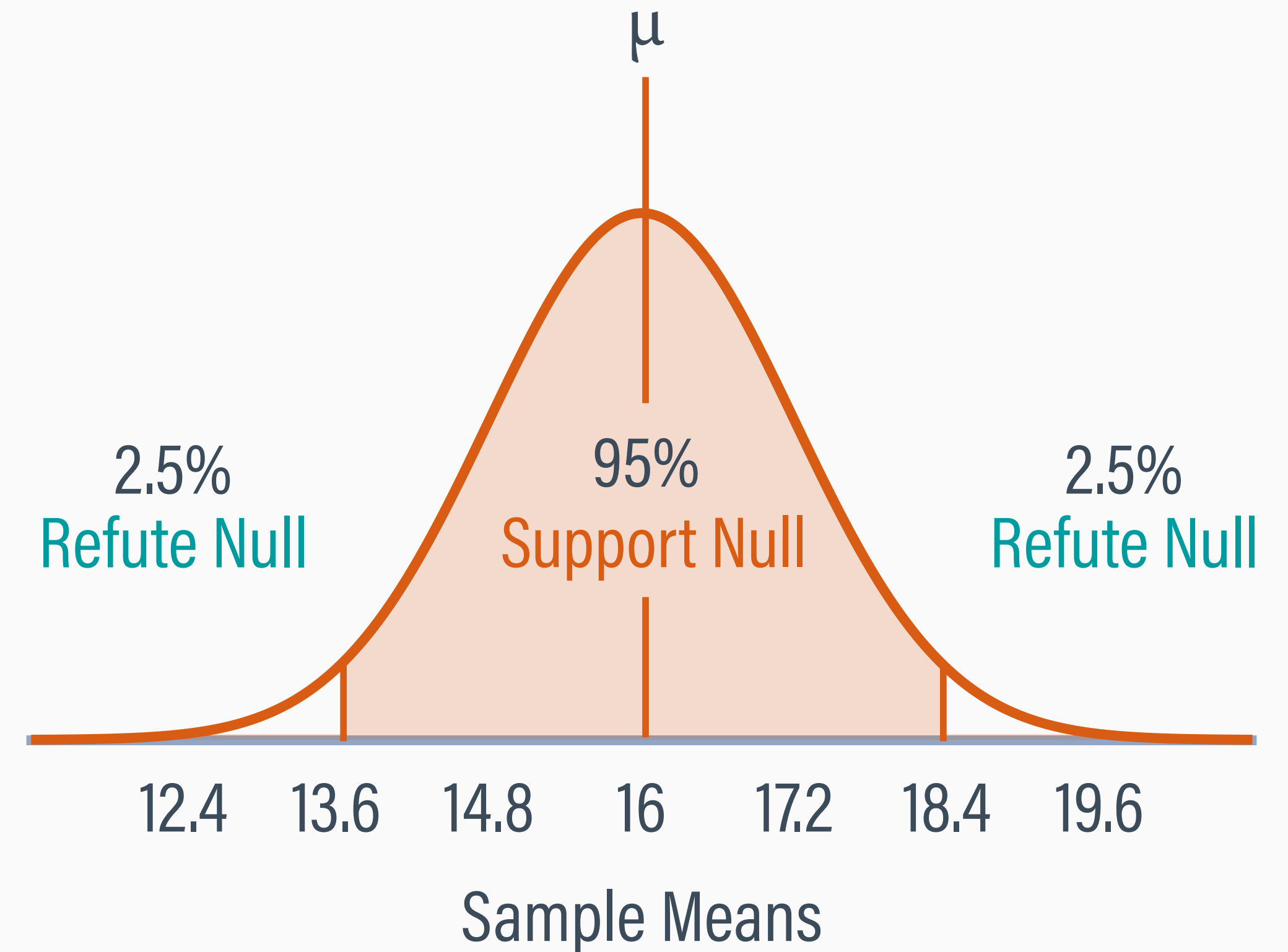
Evaluate hypotheses and draw conclusion

RESEARCH QUESTION REVISITED

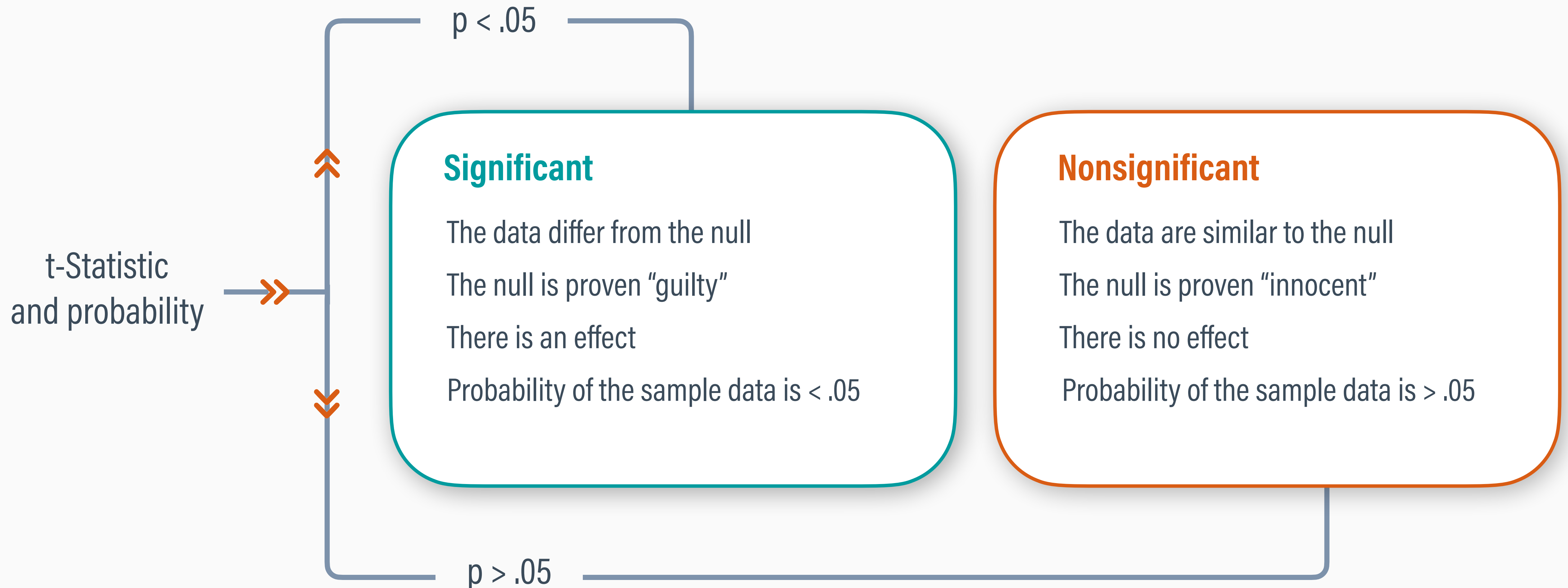
- Studies typically attempt to answer a handful of research questions involving associations between key variables
- Do people who receive a positive cancer diagnosis experience clinical levels of depressive symptoms, or do they show no meaningful elevation?
- CES-D scores > 16 are widely viewed as indicating risk for clinical depression, and scores < 16 are in the mild range

5% SIGNIFICANCE CRITERION REVISITED

- By convention, we refute the null if the sample $\bar{X}$ falls outside the middle 95% of the sampling distribution ($p < .05$)
- Such as sample has less than a 5% chance of originating from the null population
- We deem the null implausible because our data are unlikely to originate from that population

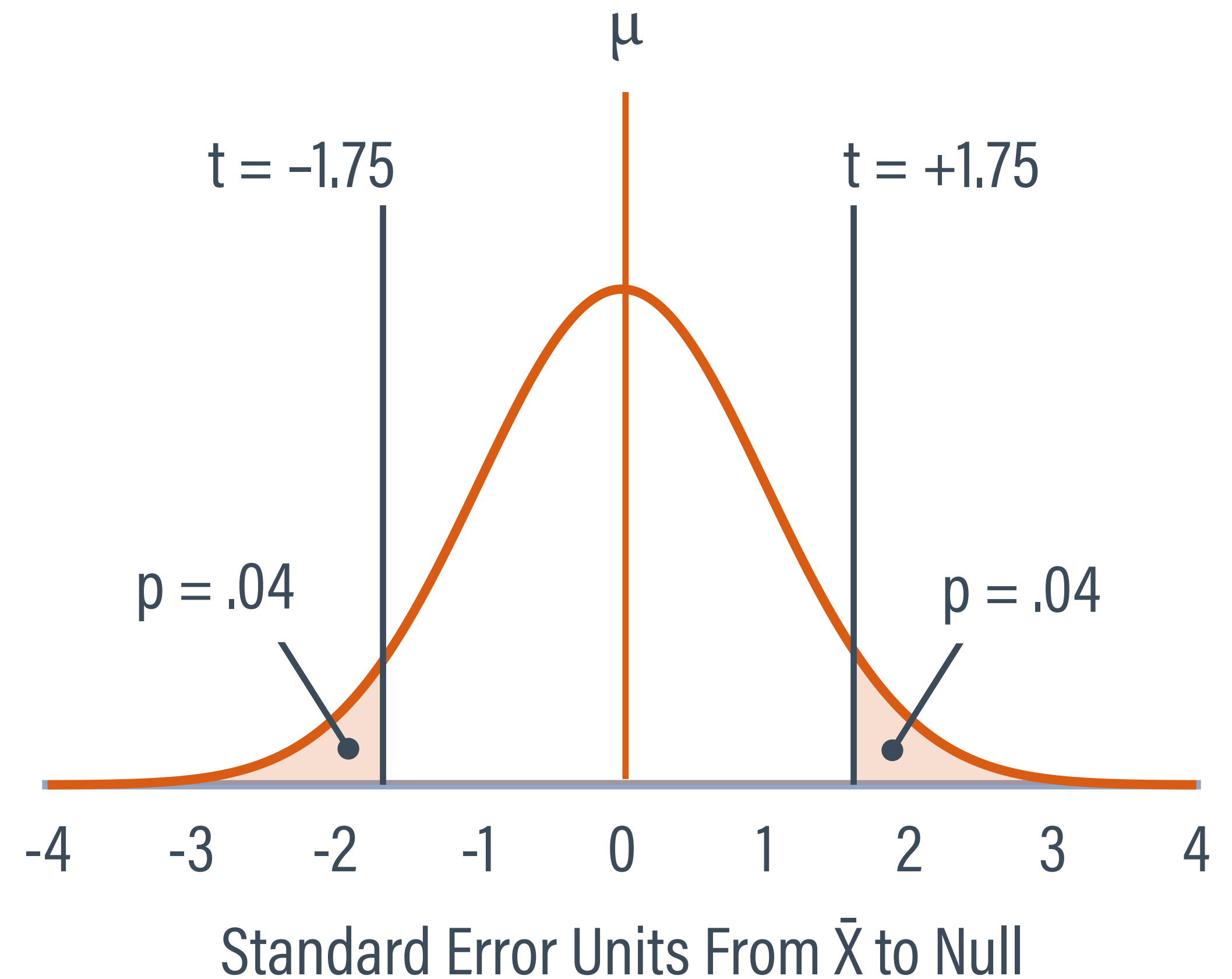


DECISION TREE





The two-tailed probability for the depression study is $p = .08$. In small groups of two or three, discuss your decision about the the null hypothesis. Translate your decision into a tangible statement about the effect of a cancer diagnosis on depressive symptoms.

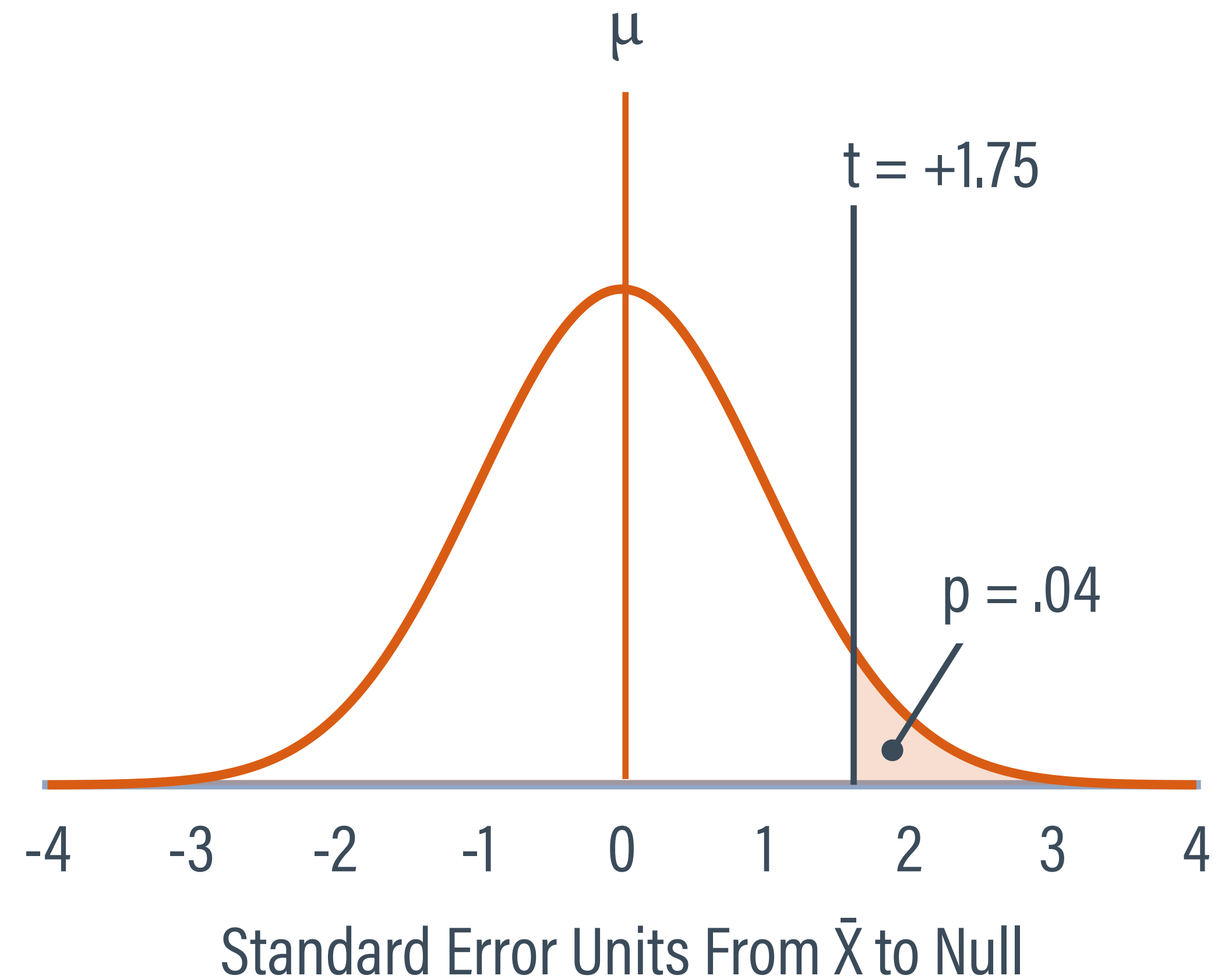


CONCLUSION: TWO-TAILED ALTERNATE

- The p-value of .08 (8%) would lead us to support the null
- A mean as large as $\bar{X} = 18.10$ could have plausibly originated from a null population with $\mu = 16$
- There is not evidence of clinical-level depression symptoms, nor is there evidence that depression is at subclinical levels



Suppose the researchers specified a one-tailed hypothesis where only an increase in depression could refute the null. The one-tailed probability for the depression study is $p = .04$. In small groups of two or three, discuss your decision about the the null hypothesis. Translate your decision into a tangible statement about the effect of a cancer diagnosis on depressive symptoms.



CONCLUSION: ONE-TAILED ALTERNATE

- The p-value of .04 (4%) would lead us to refute the null
- A mean as large as $\bar{X} = 18.10$ is unlikely to have originated from a null population with $\mu = 16$
- We have evidence that a cancer diagnosis is associated with clinically elevated depression

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STUDY QUESTIONS

Use the following research scenario to answer the study questions:

Researchers want to evaluate a new medication for smoking cessation. The outcome variable, breath carbon monoxide, is a common biomarker of smoking behavior in clinical trials. The researchers administer the medication to a sample of $N = 165$ participants, and they obtain a sample mean of $\bar{X} = 5.5$. It is widely believed that breath CO levels of 5 reflect successful outcomes in clinical trials. To evaluate the medication, you will perform significance testing steps assuming a population with a true mean of $\mu = 5$ (i.e., a population where the trial was deemed a success).

STUDY QUESTIONS (1)

1. State the null hypothesis, both as a sentence and using statistical symbols.
2. State the two-tailed alternate hypothesis, both as a sentence and using statistical symbols.

STUDY QUESTIONS (2)

3. State a one-tailed alternate hypothesis of your choosing (devise one that you think makes sense), both as a sentence and using statistical symbols.
4. Explain whether you think a one- or two-tailed alternate hypothesis is most appropriate for this research scenario.

STUDY QUESTIONS (3)

5. The sampling distribution under the null hypothesis plays a vital role in hypothesis testing. Explain how the 5% significance criterion is applied to this distribution, and how it is used to decide whether to reject the null hypothesis.
6. A sample of $N = 165$ participants had a mean of $\bar{X} = 5.5$. The test statistic was $t = +1.10$. Explain what the t-statistic measures. What do the sign and the magnitude of the t-statistic indicate about the plausibility of the null hypothesis?

STUDY QUESTIONS (4)

7. Researchers report the results as “statistically significant.” What does this imply about whether a sample mean of $\bar{X} = 5.5$ is likely to have come from a null population with $\mu = 5$?
8. Researchers report the results as “statistically non-significant.” What does this imply about whether a sample mean of $\bar{X} = 5.5$ is likely to have come from a null population with $\mu = 5$?

STUDY QUESTIONS (5)

9. The two-tailed p-value was .25. Provide an interpretation of the probability value.
10. Consider the following statement, which is a common misinterpretation about the probability value: $p = .25$ means there's a 25% chance the null is true (and 75% chance the alternative is true). Discuss why this interpretation is incorrect.
11. Still referring to the p-value of .25, what does this tell you about whether the trial could be deemed effective?

OUTLINE

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R analysis

LOAD PACKAGES AND IMPORT DATA

-  = data frame name
-  = variable name
-  = raw data file name

```
# LOAD R PACKAGES ----
```

```
# load R packages
library(ggplot2)
library(psych)
library(summarytools)
```

```
# READ DATA ----
```

```
# github url for raw data
filepath <-
  'https://raw.githubusercontent.com/craigenders/psych250a/main/data/CancerPositiveData.csv'
```

```
# create data frame called Cancer from github data
Cancer <- read.csv(filepath, stringsAsFactors = T)
```


SUMMARIZING DATA

 = data frame name
 = variable name

```
# INSPECT DATA ----
```

```
# summarize entire data frame (summarytools package)  
dfSummary()
```

```
# DESCRIPTIVE STATISTICS ----
```

```
# descriptive statistics for entire data frame (psych package)  
describe()
```

OUTPUT

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Participant	1	107	161.06	84.89	165.00	163.53	100.82	6.00	299.00	293.00	-0.13	-1.15	8.21
Diagnosis*	2	107	1.00	0.00	1.00	1.00	0.00	1.00	1.00	0.00	NaN	NaN	0.00
Age	3	107	58.96	12.84	60.00	59.40	11.86	19.00	89.00	70.00	-0.40	0.51	1.24
Gender*	4	107	1.45	0.50	1.00	1.44	0.00	1.00	2.00	1.00	0.20	-1.98	0.05
Comorbids	5	107	0.94	0.75	1.00	0.93	1.48	0.00	2.00	2.00	0.09	-1.24	0.07
Optimism	6	107	8.98	2.98	9.00	9.20	2.97	0.00	14.00	14.00	-0.66	0.00	0.29
Depression	7	107	18.10	12.45	14.00	16.55	10.38	1.00	60.00	59.00	1.12	0.80	1.20
VisImpair	8	107	5.96	2.18	5.72	5.86	2.18	2.11	12.18	10.07	0.45	-0.32	0.21

T-TESTS

 = data frame name
 = variable name

```
# T-TEST ----
```

```
# default two-tailed test with null mean = 16 (base R)
```

```
t.test($, mu = 16, alternative = 'two.sided')
```

```
# one-tailed test in the positive direction with null mean = 16 (base R)
```

```
t.test($, mu = 16, alternative = "greater")
```


OUTPUT

One Sample t-test

```
data: Cancer$Depression
t = 1.7468, df = 106, p-value = 0.08357
alternative hypothesis: true mean is not equal to 16
```

t-statistic and p-value

```
95 percent confidence interval:
15.71617 20.48944
```

95% confidence interval

```
sample estimates:
mean of x
18.1028
```

Sample mean

OUTPUT

One Sample t-test

```
data: Cancer$Depression
t = 1.7468, df = 106, p-value = 0.04178
alternative hypothesis: true mean is greater than 16
```

t-statistic and p-value

```
95 percent confidence interval:
16.10528      Inf
```

95% confidence interval

```
sample estimates:
mean of x
18.1028
```

Sample mean